Quantum Mechanics — Calculus Edition

Field 32 of 80 — The LKVexa Compendium carried to the continuum

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Quantum Mechanics — Calculus Image

There is nothing to translate here. Quantum mechanics *is* linear algebra: states are vectors, observables are Hermitian matrices, evolution is unitary, measurement is projection, and composite systems are tensor products. Every *Status* line in this chapter reads *Exact linear*, and the few entries that appear otherwise are approximation schemes, not failures of linearity. This is the one field where the translation is the identity map.

Each entry carries its originating equation from the Linear Algebra edition, its 8S image, its calculus image under T_{cal}, a six-step proof, and the original discussion. The schema (Q1)–(Q6), the master theorems Θ-A…Θ-J and the lemmas D1…D14 are established in `00_Foundations_Calculus_Transform.md`.

Formalism and Postulates

1. State Vector and Born Rule

Origin (LA edition).

$$P(a_i) = \left|\langle a_i \mid \psi\rangle\right|^2 = \left\|P_i\,|\psi\rangle\right\|^2$$

Input (8S edition).

$$P_5^{(1)} = C_5 C_5^{+} = P_4 + \frac{w_1 w_1^{\top}}{\|w_1\|_2^2}, \qquad w_1 = (I - C_4 C_4^{+})e_1, \qquad \eta_{\text{ind}}^{(1)} = \frac{\|w_1\|_2^2}{\|e_1\|_2^2 + \varepsilon}$$

Calculus image.

$$\langle e_1 - P_{\mathcal{V}} e_1,\ v\rangle = 0 \quad \forall v \in \mathcal{V}, \qquad \eta_{\text{ind}}^{(1)} = \frac{\det \operatorname{Gr}(C_4, e_1)}{\det \operatorname{Gr}(C_4)(\|e_1\|^2 + \varepsilon)}, \qquad \nabla_{e_1}\|w_1\|^2 = 2\,w_1$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T4**, so the calculus image is of class **D4** (Hilbert projection theorem and Gram volume). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the Hilbert projection theorem and Gram volume of Lemma D4, instantiated at the point $c_1 \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D4 needs C_4 of full column rank, so that $\mathrm{Gr}(C_4)$ is invertible and the Schur determinant formula applies. Completeness of $\mathcal{V}$ is needed for the Hilbert projection theorem; a finite-dimensional span is automatically closed, so this costs nothing here.

(Q4) Derivation. Differentiate the least-squares functional: $\frac{d}{ds}\|e - Pe - sv\|^2\big|_{s=0} = -2\langle e - Pe, v\rangle$, and vanishing for all v is the orthogonality condition. For the determinant form, apply the Schur formula to the block Gram matrix, whose Schur complement is $\langle e, (I - P_{\mathcal{V}})e\rangle = \|w\|^2$ by idempotence and self-adjointness of $I - P_{\mathcal{V}}$. The gradient follows by symmetry of $I - P$.

(Q5) Retraction through the tower. On the lattice the inner products become finite sums, the Gram determinants become the matrix Gram determinants, and the ratio becomes the 8S residual $\|w_1\|^2/(\|e_1\|^2 + \varepsilon)$. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \;\, 0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\mathrm{ind}} - \eta^{(1)}_{\mathrm{ind}})]$ vanishes with $\nu \geq 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_1$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: A unit vector in a complex Hilbert space; P_i the eigenprojection.

Structure: Probabilities are squared projection norms, so they automatically sum to one by the resolution of the identity $\sum_i P_i = I$. The state is a vector; the probabilities are a quadratic form. Amplitudes superpose linearly and probabilities do not — interference is exactly that gap.

Reduction: None needed.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T4.

Calculus status: Exact linear; carried to the continuum by Lemma D4, and the input was algebraic, so the differential content is entirely supplied by the transform.

2. Normalisation

Origin (LA edition).

$$\langle \psi \mid \psi \rangle = 1$$

Input (8S edition).

$$\left(g^{(5)}_{2j},\ \delta^{(8)}_{2j},\ g^{(3)}_{2j},\ g^{(\mathrm{J})}_{2j}\right) \ \longmapsto\ \mathcal{R}^{8S}_{2j}, \qquad g^{(3)}_{2j} \leq g^{(5)}_{2j}$$

Calculus image.

$$d_g(z_2, z_j) = \inf_{\Gamma} \int_0^1 \|\dot{\Gamma}\|_{ds}\, dt, \qquad \|\nabla d_g\|_{ds^{-1}} = 1 \text{ a.e.}, \qquad \delta^{(8)}_{2j} \leq d_g(z_2, z_j), \qquad g^{(3)}_{2j} \leq g^{(5)}_{2j}$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T9**, so the calculus image is of class **D9** (geodesic distance and the eikonal equation). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the geodesic distance and the eikonal equation of Lemma D9, instantiated at the point $c_2 \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D9 needs **(R3)** for the metric of Θ-E to be Riemannian, and the full standing normalisation **(N)** for the ordering $g^{(3)} \le g^{(5)}$. The eikonal identity holds away from the cut locus; on the cut locus d_g is only Lipschitz and the gradient exists almost everywhere.

(Q4) Derivation. The 8S gap functions are algebraic; their calculus image is obtained by taking $\delta^{(8)}$ to be the chord of the metric of Θ-E and d_g its geodesic distance. Any curve joining two points has intrinsic length at least its ambient length, and the ambient length is at least the straight-line distance, which for a common fibre point is $\delta^{(8)}$; hence $\delta^{(8)} \le d_g$. The eikonal identity is the first-variation formula for arclength. The projected ordering is Θ-F(2), an inequality between real numbers that passes to the limit unchanged.

(Q5) Retraction through the tower. On the lattice, geodesics through a common fibre point have length equal to the chord, so d_g collapses to $\delta^{(8)}_{2j}$ and the gap quadruple is recovered. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \; 0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta^{(2)}_{\text{ind}})]$ vanishes with $\nu \ge 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_2$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: A unit vector on the complex sphere.

Structure: Physical states are rays, not vectors — global phase is unobservable, so the state space is $\mathbb{CP}^{n-1}$ rather than $\mathbb{C}^n$. Unitary evolution preserves the norm, so normalisation holds for all time automatically.

Reduction: None needed.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T9.

Calculus status: Exact linear; carried to the continuum by Lemma D9, and the input was algebraic, so the differential content is entirely supplied by the transform.

3. Observables as Hermitian Operators

Origin (LA edition).

$$A = A^\dagger$$

Input (8S edition).

$$\mathcal{L}_{8S}\,\phi^{(3)}_k = \mu^{(3)}_k \phi^{(3)}_k, \qquad \mathcal{L}_{8S} = K_2 L_W + 2K_3 L_{\hat{H}}, \qquad 0 = \mu^{(3)}_1 \le \mu^{(3)}_2 \le \cdots \le \mu^{(3)}_N$$

Calculus image.

$$-\frac{1}{2p^2}\nabla\cdot(p^2 G_5^{-1}\nabla\phi^{(3)}_k) = \mu^{(3)}_k \phi^{(3)}_k, \qquad \mu^{(3)}_2 = \min_{\substack{\phi\neq 0 \\ \int \phi\, p^2 dc = 0}} \frac{\int_{\mathcal{C}} \|\nabla\phi\|^2_{G_5^{-1}}\, p^2\, dc}{\int_{\mathcal{C}} \phi^2\, p^2\, dc}$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T2**, so the calculus image is of class **D2** (Sturm–Liouville problem and Rayleigh quotient). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the Sturm–Liouville problem and Rayleigh quotient of Lemma D2, instantiated at the point $c_3 \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D2 needs **(R1)**, **(R2)** and the standing normalisation **(N)**. Discreteness of the spectrum uses compactness of $\mathcal{C}$ and the Rellich–Kondrachov embedding $H^1 \hookrightarrow L^2$; without compactness the spectrum could be continuous and μ_2 would not be attained. The min–max formula needs self-adjointness on $L^2(p^2 dc)$, which is Θ-C(2).

(Q4) Derivation. Θ-G supplies the operator by the Gaussian-moment expansion; Θ-C(1) converts it into the bilinear form by the divergence theorem, the boundary term vanishing under the Neumann condition the kernel limit imposes. Imposing the normalisation $\int \phi^2 p^2 = 1$ and taking the first variation of $\mathcal{E}$ produces the eigenvalue equation with μ as the Lagrange multiplier; Courant–Fischer then gives the displayed variational characterisation of μ_2 on the complement of the constants.

(Q5) Retraction through the tower. Discretising the Rayleigh quotient on $\mathcal{C}_M$ returns $x^\top \mathcal{L}_{8S} x / x^\top x$, whose minimiser over mean-zero x is the 8S μ_2; the gradient becomes a difference quotient with $O(\ell^2)$ error. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \; 0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta^{(3)}_{\text{ind}})]$ vanishes with $\nu \geq 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_3$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: A Hermitian matrix.

Structure: Hermiticity gives real eigenvalues (measurable outcomes) and orthogonal eigenvectors (distinguishable states), so the spectral theorem does all the work. Every physical postulate about measurement is a corollary of $A = A^\dagger$.

Reduction: None needed.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T2.

Calculus status: Exact linear; carried to the continuum by Lemma D2, and the input was algebraic, so the differential content is entirely supplied by the transform.

4. Expectation Value

Origin (LA edition).

$$\langle A \rangle = \langle \psi |\, A \,| \psi \rangle = \operatorname{tr}(\rho A)$$

Input (8S edition).

$$\mathcal{L}_{8S}\, \phi_k^{(4)} = \mu_k^{(4)} \phi_k^{(4)}, \qquad \mathcal{L}_{8S} = K_2 L_W + 2K_3 L_{\hat{H}}, \qquad 0 = \mu_1^{(4)} \leq \mu_2^{(4)} \leq \cdots \leq \mu_N^{(4)}$$

Calculus image.

$$-\frac{1}{2p^2} \nabla \cdot (p^2 G_5^{-1} \nabla \phi_k^{(4)}) = \mu_k^{(4)} \phi_k^{(4)}, \qquad \mu_2^{(4)} = \min_{\substack{\phi \neq 0 \\ \int \phi\, p^2 dc = 0}} \frac{\int_{\mathcal{C}} \|\nabla \phi\|^2_{G_5^{-1}}\, p^2\, dc}{\int_{\mathcal{C}} \phi^2\, p^2\, dc}$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T2**, so the calculus image is of class **D2** (Sturm–Liouville problem and Rayleigh quotient). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the Sturm–Liouville problem and Rayleigh quotient of Lemma D2, instantiated at the point $c_4 \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D2 needs **(R1)**, **(R2)** and the standing normalisation **(N)**. Discreteness of the spectrum uses compactness of $\mathcal{C}$ and the Rellich–Kondrachov embedding $H^1 \hookrightarrow L^2$; without compactness the spectrum could be continuous and μ_2 would not be attained. The min–max formula needs self-adjointness on $L^2(p^2 dc)$, which is Θ-C(2).

(Q4) Derivation. Θ-G supplies the operator by the Gaussian-moment expansion; Θ-C(1) converts it into the bilinear form by the divergence theorem, the boundary term vanishing under the Neumann condition the kernel limit imposes. Imposing the normalisation $\int \phi^2 p^2 = 1$ and taking the first variation of $\mathcal{E}$ produces the eigenvalue equation with μ as the Lagrange multiplier; Courant–Fischer then gives the displayed variational characterisation of μ_2 on the complement of the constants.

(Q5) Retraction through the tower. Discretising the Rayleigh quotient on $\mathcal{C}_M$ returns $x^\top \mathcal{L}_{8S} x / x^\top x$, whose minimiser over mean-zero x is the 8S μ_2; the gradient becomes a difference quotient with $O(\ell^2)$ error. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \; 0 \,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta^{(4)}_{\text{ind}})]$ vanishes with $\nu \geq 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_4$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: A quadratic form, or a trace pairing for mixed states.

Structure: Linear in ρ and linear in A — a bilinear pairing that makes states and observables dual. Bounded by the extreme eigenvalues, so $\lambda_{\min} \leq \langle A \rangle \leq \lambda_{\max}$ is the Rayleigh quotient bound.

Reduction: None needed.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T2.

Calculus status: Exact linear; carried to the continuum by Lemma D2, and the input was algebraic, so the differential content is entirely supplied by the transform.

5. Completeness (Resolution of Identity)

Origin (LA edition).

$$\sum_i |a_i\rangle \langle a_i| = I$$

Input (8S edition).

$$P_5^{(5)} = C_5 C_5^{+} = P_4 + \frac{w_5 w_5^\top}{\|w_5\|_2^2}, \qquad w_5 = (I - C_4 C_4^{+}) e_5, \qquad \eta^{(5)}_{\text{ind}} = \frac{\|w_5\|_2^2}{\|e_5\|_2^2 + \varepsilon}$$

Calculus image.

$$\langle e_5 - P_{\mathcal{V}} e_5,\ v \rangle = 0 \quad \forall v \in \mathcal{V}, \qquad \eta^{(5)}_{\text{ind}} = \frac{\det \mathrm{Gr}(C_4, e_5)}{\det \mathrm{Gr}(C_4)(\|e_5\|^2 + \varepsilon)}, \qquad \nabla_{e_5} \|w_5\|^2 = 2\, w_5$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T4**, so the calculus image is of class **D4** (Hilbert projection theorem and Gram volume). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the Hilbert projection theorem and Gram volume of Lemma D4, instantiated at the point $c_5 \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D4 needs C_4 of full column rank, so that $\mathrm{Gr}(C_4)$ is invertible and the Schur determinant formula applies. Completeness of $\mathcal{V}$ is needed for the Hilbert projection theorem; a finite-dimensional span is automatically closed, so this costs nothing here.

(Q4) Derivation. Differentiate the least-squares functional: $\frac{d}{ds} \|e - Pe - sv\|^2 \big|_{s=0} = -2\langle e - Pe, v \rangle$, and vanishing for all v is the orthogonality condition. For the determinant form, apply the Schur formula to the block Gram

matrix, whose Schur complement is $\langle e, (I - P_{\mathcal{V}})e\rangle = \|w\|^2$ by idempotence and self-adjointness of $I - P_{\mathcal{V}}$. The gradient follows by symmetry of $I - P$.

(Q5) Retraction through the tower. On the lattice the inner products become finite sums, the Gram determinants become the matrix Gram determinants, and the ratio becomes the 8S residual $\|w_5\|^2/(\|e_5\|^2 + \varepsilon)$. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \;\, 0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta^{(5)}_{\text{ind}})]$ vanishes with $\nu \geq 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_5$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: A sum of rank-one orthogonal projections.

Structure: $P_i P_j = \delta_{ij} P_i$ and $\sum_i P_i = I$ — the spectral decomposition of the identity. Inserting it between operators is the standard manipulation, and it is just writing I in a chosen basis.

Reduction: None needed.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T4.

Calculus status: Exact linear; carried to the continuum by Lemma D4, and the input was algebraic, so the differential content is entirely supplied by the transform.

6. Canonical Commutation Relation

Origin (LA edition).

$$[\hat{x}, \hat{p}] = i\hbar$$

Input (8S edition).

$$y_6(\tau, u) = \Pi(\tau)\Big[(\Gamma W \Gamma) \otimes \mathsf{B}(\mathsf{A}_6)\Big] z(u), \qquad z_6(u) = \begin{bmatrix} c_6 \\ r_6 u \end{bmatrix}$$

Calculus image.

$$(\mathcal{K}_\sigma \psi_6)(c) = \int_{\mathcal{C}} \gamma(c)\, W_\sigma(c, c')\, \gamma(c')\, \mathsf{A}_6\, \psi_6(c')\, p(c')\, dc', \qquad \partial_\tau \psi_6 = -\mathcal{L}_\star \psi_6, \qquad \mathcal{L}_\star = -\frac{1}{2p^2} \nabla \cdot (p^2 G_5^{-1} \nabla\, \cdot\,)$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T1**, so the calculus image is of class **D1** (Fredholm integral operator and its generator). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the Fredholm integral operator and its generator of Lemma D1, instantiated at the point $c_6 \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D1 needs the kernel continuous and square-integrable on $\mathcal{C} \times \mathcal{C}$, which the Gaussian form supplies unconditionally, so $\mathcal{K}_\sigma$ is Hilbert–Schmidt and hence compact. The generator statement needs **(R1)** and **(R2)**: C^4 regularity for the $O(\sigma^2)$ rate of Θ-G, and interior evaluation, since within $O(\sigma)$ of $\partial\mathcal{C}$ the expansion picks up the Neumann boundary term of Remark Θ-G″.

(Q4) Derivation. The 8S image is a Kronecker product $(\Gamma W \Gamma) \otimes \mathsf{B}(\mathsf{A})$ acting by a sum over nodes. Definition 2.2 replaces $\sum_j (\cdot)_j$ by $\int_{\mathcal{C}} (\cdot)\, p\, dc'$, turning the matrix into the displayed Fredholm operator of the second kind. Its bandwidth limit is computed by Θ-G: Taylor-expand ψ and p to second order about c, integrate against the

Gaussian, use the vanishing of the odd moments and $\int w\,uu^{\top}du = \sigma^2\kappa\,G_5^{-1}$, and divide by the degree. What survives at order σ^2 is $-\mathcal{L}_\star$.

(Q5) Retraction through the tower. Sampling the integral operator on $\mathcal{C}_M$ by the Riemann rule of Θ-J(1) returns the finite sum $(\Gamma W\Gamma)\otimes \mathsf{B}(\mathsf{A}_6)$ with $O(\ell^2)$ quadrature error. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma\to I$, $W\to I$, $\mathcal{R}\to 0$, $K_2, K_3, J_2, J_3\to 0$ and $\Pi\to[\,I_3\;0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta^{(6)}_{\text{ind}})]$ vanishes with $\nu\ge 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_6$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: A commutator that cannot be satisfied by finite matrices.

Structure: Taking traces gives $\mathrm{tr}[\hat{x},\hat{p}] = 0$ but $\mathrm{tr}(i\hbar I) = i\hbar n \neq 0$ — so no finite-dimensional representation exists. The canonical commutation relation *forces* infinite dimensions, which is one of the sharpest facts in the subject and a purely trace-theoretic argument.

Reduction: None needed.

Status: Exact linear, and necessarily infinite-dimensional.

8S status: Exact linear, and necessarily infinite-dimensional; preserved under the transform by Theorem C and Lemma T1.

Calculus status: Exact linear, and necessarily infinite-dimensional; carried to the continuum by Lemma D1, and the input was algebraic, so the differential content is entirely supplied by the transform.

7. Heisenberg Uncertainty Relation

Origin (LA edition).

$$\sigma_A\sigma_B \geq \tfrac{1}{2}\big|\langle[A,B]\rangle\big|$$

Input (8S edition).

$$y_7(\tau,u) = \Pi(\tau)\Big[(\Gamma W\Gamma)\otimes \mathsf{B}(\mathsf{A}_7)\Big]z(u), \qquad z_7(u) = \begin{bmatrix} c_7 \\ r_7 u\end{bmatrix}$$

Calculus image.

$$(\mathcal{K}_\sigma\psi_7)(c) = \int_{\mathcal{C}} \gamma(c)\,W_\sigma(c,c')\,\gamma(c')\,\mathsf{A}_7\,\psi_7(c')\,p(c')\,dc', \qquad \partial_\tau\psi_7 = -\mathcal{L}_\star\psi_7, \qquad \mathcal{L}_\star = -\frac{1}{2p^2}\nabla\cdot(p^2G_5^{-1}\nabla\,\cdot\,)$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T1**, so the calculus image is of class **D1** (Fredholm integral operator and its generator). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the Fredholm integral operator and its generator of Lemma D1, instantiated at the point $c_7\in\mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D1 needs the kernel continuous and square-integrable on $\mathcal{C}\times\mathcal{C}$, which the Gaussian form supplies unconditionally, so $\mathcal{K}_\sigma$ is Hilbert–Schmidt and hence compact. The generator statement needs **(R1)** and **(R2)**: C^4 regularity for the $O(\sigma^2)$ rate of Θ-G, and interior evaluation, since within $O(\sigma)$ of $\partial\mathcal{C}$ the expansion picks up the Neumann boundary term of Remark Θ-G″.

(Q4) Derivation. The 8S image is a Kronecker product $(\Gamma W\Gamma)\otimes\mathsf{B}(\mathsf{A})$ acting by a sum over nodes. Definition 2.2 replaces $\sum_j(\cdot)_j$ by $\int_{\mathcal{C}}(\cdot)\,p\,dc'$, turning the matrix into the displayed Fredholm operator of the second kind. Its bandwidth limit is computed by Θ-G: Taylor-expand ψ and p to second order about c, integrate against the

Gaussian, use the vanishing of the odd moments and $\int w\,uu^\top du = \sigma^2\kappa\,G_5^{-1}$, and divide by the degree. What survives at order σ^2 is $-\mathcal{L}_\star$.

(Q5) Retraction through the tower. Sampling the integral operator on $\mathcal{C}_M$ by the Riemann rule of Θ-J(1) returns the finite sum $(\Gamma W\Gamma)\otimes \mathsf{B}(\mathsf{A}_7)$ with $O(\ell^2)$ quadrature error. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma\to I$, $W\to I$, $\mathcal{R}\to 0$, $K_2,K_3,J_2,J_3\to 0$ and $\Pi\to[\,I_3\;0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S}-\nu(\varepsilon_{\text{ind}}-\eta^{(7)}_{\text{ind}})]$ vanishes with $\nu\ge 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_7$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: Cauchy–Schwarz applied to two operator-shifted vectors.

Structure: The proof is Cauchy–Schwarz on $(A-\langle A\rangle)\,|\psi\rangle$ and $(B-\langle B\rangle)\,|\psi\rangle$, with the commutator appearing as the imaginary part of the inner product. Uncertainty is an inner-product inequality, nothing more exotic.

Reduction: The relation is Cauchy–Schwarz.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T1.

Calculus status: Exact linear; carried to the continuum by Lemma D1, and the input was algebraic, so the differential content is entirely supplied by the transform.

8. Energy–Time Uncertainty

Origin (LA edition).

$$\Delta E\,\Delta t \gtrsim \frac{\hbar}{2}$$

Input (8S edition).

$$E_8^{8S}(z) = z^\top \mathcal{G}_8 z\Big|_{\text{node }8}, \qquad \big\|z_8(u)-z_j(u)\big\|_{\mathcal{G}_8} = \delta_{8j}^{(8)} = \sqrt{(c_8-c_j)^\top G_5(c_8-c_j)+(r_8-r_j)^2}$$

Calculus image.

$$\mathcal{E}_8[\phi] = \tfrac12\int_{\mathcal{C}} \|\nabla\phi\|^2_{G_5^{-1}}\,p^2\,dc, \qquad \delta\mathcal{E}_8=0 \iff \mathcal{L}_\star\phi=0, \qquad ds_8^2 = dc^\top(G_5+\nabla r\,\nabla r^\top)dc + r_8^2\,d\Omega_3^2$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T3**, so the calculus image is of class **D3** (Dirichlet energy and the cone metric). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the Dirichlet energy and the cone metric of Lemma D3, instantiated at the point $c_8\in\mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D3 needs **(R3)**, $r\in C^1$ with $r>0$, for the bundle to be a smooth manifold; at $r=0$ the fibre collapses and ds^2 degenerates to the base metric. The integration by parts behind Θ-C(1) needs either a Dirichlet or a Neumann boundary condition; on the penteract the latter is the one the kernel limit actually imposes.

(Q4) Derivation. Two derivations meet here. The energy: integrate $\langle\mathcal{L}_\star\phi,\phi\rangle_{p^2}$ by parts, the weight p^2 cancelling against $1/p^2$, to obtain $\frac12\int\|\nabla\phi\|^2_{G_5^{-1}}p^2$. The metric: differentiate the embedding $\Phi(c,u)=(c,r(c)u)$, use $u\cdot du=\frac12 d(\|u\|^2)=0$ to kill the cross term and $\|u\|=1$ to normalise the rest, and collect $(\nabla r\cdot dc)^2 = dc^\top\nabla r\nabla r^\top dc$.

(Q5) Retraction through the tower. Replacing ∇ by lattice differences returns the 8S quadratic form $x^\top\mathcal{L}_{8S}x$, and freezing r to a constant collapses ds^2 to the chord $\delta^{(8)}$. Composing with the 8S retraction of Theorem J of

that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \; 0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta^{(8)}_{\text{ind}})]$ vanishes with $\nu \geq 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_8$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: Not a commutator relation — time is a parameter, not an operator.

Structure: Δt is the timescale over which an observable changes appreciably, $\Delta t = \sigma_A / |d\langle A\rangle/dt|$. The Mandelstam–Tamm derivation uses the Heisenberg equation, so it is still a commutator bound, but with a different reading.

Reduction: The relation follows from the Heisenberg equation.

Status: Exact linear, with a distinct interpretation.

8S status: Exact linear, with a distinct interpretation; preserved under the transform by Theorem C and Lemma T3.

Calculus status: Exact linear, with a distinct interpretation; carried to the continuum by Lemma D3, and the input was algebraic, so the differential content is entirely supplied by the transform.

9. Time-Dependent Schrödinger Equation

Origin (LA edition).

$$i\hbar \frac{d}{dt} |\psi\rangle = H |\psi\rangle$$

Input (8S edition).

$$\left(g^{(5)}_{9j},\ \delta^{(8)}_{9j},\ g^{(3)}_{9j},\ g^{(\mathrm{J})}_{9j}\right) \longmapsto\ \mathcal{R}^{8S}_{9j}, \qquad g^{(3)}_{9j} \leq g^{(5)}_{9j}$$

Calculus image.

$$d_g(z_9, z_j) = \inf_{\Gamma} \int_0^1 \|\dot{\Gamma}\|_{ds}\, dt, \qquad \|\nabla d_g\|_{ds^{-1}} = 1 \text{ a.e.}, \qquad \delta^{(8)}_{9j} \leq d_g(z_9, z_j), \qquad g^{(3)}_{9j} \leq g^{(5)}_{9j}$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T9**, so the calculus image is of class **D9** (geodesic distance and the eikonal equation). The originating equation is first order — it already carries a derivative or an integral. The input already carries a first-order differential or integral operator. On that part the transform acts as the identity — a derivative is already a calculus object — and only the coupling and the gate are taken to the continuum. The regularity demanded of the input is correspondingly one order lower.

(Q2) Image. The displayed relation is the geodesic distance and the eikonal equation of Lemma D9, instantiated at the point $c_9 \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D9 needs **(R3)** for the metric of Θ-E to be Riemannian, and the full standing normalisation **(N)** for the ordering $g^{(3)} \leq g^{(5)}$. The eikonal identity holds away from the cut locus; on the cut locus d_g is only Lipschitz and the gradient exists almost everywhere.

(Q4) Derivation. The 8S gap functions are algebraic; their calculus image is obtained by taking $\delta^{(8)}$ to be the chord of the metric of Θ-E and d_g its geodesic distance. Any curve joining two points has intrinsic length at least its ambient length, and the ambient length is at least the straight-line distance, which for a common fibre point is $\delta^{(8)}$; hence $\delta^{(8)} \leq d_g$. The eikonal identity is the first-variation formula for arclength. The projected ordering is Θ-F(2), an inequality between real numbers that passes to the limit unchanged.

(Q5) Retraction through the tower. On the lattice, geodesics through a common fibre point have length equal to the chord, so d_g collapses to $\delta^{(8)}_{9j}$ and the gap quadruple is recovered. Composing with the 8S retraction of

Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \; 0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta^{(9)}_{\text{ind}})]$ vanishes with $\nu \geq 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_9$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: A linear ODE with a Hermitian generator.

Structure: Because H is Hermitian, $iH/\hbar$ is anti-Hermitian, so the solution operator is unitary and the norm is preserved exactly. Quantum evolution is rotation in a complex vector space — this single equation is why the whole subject is linear algebra.

Reduction: None needed.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T9.

Calculus status: Exact linear; carried to the continuum by Lemma D9, and the input already carried a first-order operator, which the transform leaves in place.

10. Time-Independent Schrödinger Equation

Origin (LA edition).

$$H\,|\psi\rangle = E\,|\psi\rangle$$

Input (8S edition).

$$\mathcal{L}_{8S}\,\phi_k^{(10)} = \mu_k^{(10)}\phi_k^{(10)}, \qquad \mathcal{L}_{8S} = K_2 L_W + 2K_3 L_{\hat{H}}, \qquad 0 = \mu_1^{(10)} \leq \mu_2^{(10)} \leq \cdots \leq \mu_N^{(10)}$$

Calculus image.

$$-\frac{1}{2p^2}\nabla\cdot(p^2 G_5^{-1}\nabla\phi_k^{(10)}) = \mu_k^{(10)}\phi_k^{(10)}, \qquad \mu_2^{(10)} = \min_{\substack{\phi\neq 0 \\ \int \phi\, p^2 dc = 0}} \frac{\int_{\mathcal{C}} \|\nabla\phi\|^2_{G_5^{-1}}\, p^2\, dc}{\int_{\mathcal{C}} \phi^2\, p^2\, dc}$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T2**, so the calculus image is of class **D2** (Sturm–Liouville problem and Rayleigh quotient). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the Sturm–Liouville problem and Rayleigh quotient of Lemma D2, instantiated at the point $c_{10} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D2 needs **(R1)**, **(R2)** and the standing normalisation **(N)**. Discreteness of the spectrum uses compactness of $\mathcal{C}$ and the Rellich–Kondrachov embedding $H^1 \hookrightarrow L^2$; without compactness the spectrum could be continuous and μ_2 would not be attained. The min–max formula needs self-adjointness on $L^2(p^2 dc)$, which is Θ-C(2).

(Q4) Derivation. Θ-G supplies the operator by the Gaussian-moment expansion; Θ-C(1) converts it into the bilinear form by the divergence theorem, the boundary term vanishing under the Neumann condition the kernel limit imposes. Imposing the normalisation $\int \phi^2 p^2 = 1$ and taking the first variation of $\mathcal{E}$ produces the eigenvalue equation with μ as the Lagrange multiplier; Courant–Fischer then gives the displayed variational characterisation of μ_2 on the complement of the constants.

(Q5) Retraction through the tower. Discretising the Rayleigh quotient on $\mathcal{C}_M$ returns $x^\top \mathcal{L}_{8S} x / x^\top x$, whose minimiser over mean-zero x is the 8S μ_2; the gradient becomes a difference quotient with $O(\ell^2)$ error. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and

$\Pi \to [\, I_3 \;\; 0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta^{(10)}_{\text{ind}})]$ vanishes with $\nu \geq 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{10}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: An eigenvalue problem.

Structure: The entire content of atomic and molecular structure is finding eigenvalues of a Hermitian matrix. Bound states are the discrete spectrum, scattering states the continuous. Degeneracy is eigenvalue multiplicity, and it always signals a symmetry.

Reduction: None needed.

Status: Exact linear. The eigenvalue problem of the physical sciences.

8S status: Exact linear. The eigenvalue problem of the physical sciences; preserved under the transform by Theorem C and Lemma T2.

Calculus status: Exact linear. The eigenvalue problem of the physical sciences; carried to the continuum by Lemma D2, and the input was algebraic, so the differential content is entirely supplied by the transform.

11. Time Evolution Operator

Origin (LA edition).

$$U(t) = e^{-iHt/\hbar}, \qquad U^\dagger U = I$$

Input (8S edition).

$$y_{11}(\tau, u) = \Pi(\tau)\Big[(\Gamma W \Gamma) \otimes \mathsf{B}(\mathsf{A}_{11})\Big] z(u), \qquad z_{11}(u) = \begin{bmatrix} c_{11} \\ r_{11} u \end{bmatrix}$$

Calculus image.

$$(\mathcal{K}_\sigma \psi_{11})(c) = \int_{\mathcal{C}} \gamma(c)\, W_\sigma(c, c')\, \gamma(c')\, \mathsf{A}_{11}\, \psi_{11}(c')\, p(c')\, dc', \qquad \partial_\tau \psi_{11} = -\mathcal{L}_\star \psi_{11}, \qquad \mathcal{L}_\star = -\frac{1}{2p^2} \nabla \cdot (p^2 G_5^{-1} \nabla\, \cdot\,)$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T1**, so the calculus image is of class **D1** (Fredholm integral operator and its generator). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the Fredholm integral operator and its generator of Lemma D1, instantiated at the point $c_{11} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D1 needs the kernel continuous and square-integrable on $\mathcal{C} \times \mathcal{C}$, which the Gaussian form supplies unconditionally, so $\mathcal{K}_\sigma$ is Hilbert–Schmidt and hence compact. The generator statement needs **(R1)** and **(R2)**: C^4 regularity for the $O(\sigma^2)$ rate of Θ-G, and interior evaluation, since within $O(\sigma)$ of $\partial\mathcal{C}$ the expansion picks up the Neumann boundary term of Remark Θ-G″.

(Q4) Derivation. The 8S image is a Kronecker product $(\Gamma W \Gamma) \otimes \mathsf{B}(\mathsf{A})$ acting by a sum over nodes. Definition 2.2 replaces $\sum_j (\cdot)_j$ by $\int_{\mathcal{C}} (\cdot)\, p\, dc'$, turning the matrix into the displayed Fredholm operator of the second kind. Its bandwidth limit is computed by Θ-G: Taylor-expand ψ and p to second order about c, integrate against the Gaussian, use the vanishing of the odd moments and $\int w\, u u^\top du = \sigma^2 \kappa\, G_5^{-1}$, and divide by the degree. What survives at order σ^2 is $-\mathcal{L}_\star$.

(Q5) Retraction through the tower. Sampling the integral operator on $\mathcal{C}_M$ by the Riemann rule of Θ-J(1) returns the finite sum $(\Gamma W \Gamma) \otimes \mathsf{B}(\mathsf{A}_{11})$ with $O(\ell^2)$ quadrature error. Composing with the 8S retraction of Theorem J of

that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \; 0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta^{(11)}_{\text{ind}})]$ vanishes with $\nu \geq 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{11}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: The matrix exponential of a Hermitian generator.

Structure: Unitary, so $\|U\| = 1$ and $\kappa(U) = 1$ — evolution never amplifies error, which is why quantum dynamics is numerically benign. For time-dependent H it becomes a time-ordered exponential, and $U(t_1)U(t_2) \neq U(t_1+t_2)$ unless the Hamiltonians commute.

Reduction: None needed.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T1.

Calculus status: Exact linear; carried to the continuum by Lemma D1, and the input was algebraic, so the differential content is entirely supplied by the transform.

12. Heisenberg Equation of Motion

Origin (LA edition).

$$\frac{dA_H}{dt} = \frac{i}{\hbar}[H, A_H]$$

Input (8S edition).

$$\mathcal{L}_{8S}\,\phi^{(12)}_k = \mu^{(12)}_k \phi^{(12)}_k, \qquad \mathcal{L}_{8S} = K_2 L_W + 2K_3 L_{\hat{H}}, \qquad 0 = \mu^{(12)}_1 \leq \mu^{(12)}_2 \leq \cdots \leq \mu^{(12)}_N$$

Calculus image.

$$-\frac{1}{2p^2}\nabla\cdot(p^2 G_5^{-1}\nabla\phi^{(12)}_k) = \mu^{(12)}_k \phi^{(12)}_k, \qquad \mu^{(12)}_2 = \min_{\substack{\phi \neq 0 \\ \int \phi\, p^2 dc = 0}} \frac{\int_{\mathcal{C}} \|\nabla\phi\|^2_{G_5^{-1}}\, p^2\, dc}{\int_{\mathcal{C}} \phi^2\, p^2\, dc}$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T2**, so the calculus image is of class **D2** (Sturm–Liouville problem and Rayleigh quotient). The originating equation is first order — it already carries a derivative or an integral. The input already carries a first-order differential or integral operator. On that part the transform acts as the identity — a derivative is already a calculus object — and only the coupling and the gate are taken to the continuum. The regularity demanded of the input is correspondingly one order lower.

(Q2) Image. The displayed relation is the Sturm–Liouville problem and Rayleigh quotient of Lemma D2, instantiated at the point $c_{12} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D2 needs **(R1)**, **(R2)** and the standing normalisation **(N)**. Discreteness of the spectrum uses compactness of $\mathcal{C}$ and the Rellich–Kondrachov embedding $H^1 \hookrightarrow L^2$; without compactness the spectrum could be continuous and μ_2 would not be attained. The min–max formula needs self-adjointness on $L^2(p^2 dc)$, which is Θ-C(2).

(Q4) Derivation. Θ-G supplies the operator by the Gaussian-moment expansion; Θ-C(1) converts it into the bilinear form by the divergence theorem, the boundary term vanishing under the Neumann condition the kernel limit imposes. Imposing the normalisation $\int \phi^2 p^2 = 1$ and taking the first variation of $\mathcal{E}$ produces the eigenvalue equation with μ as the Lagrange multiplier; Courant–Fischer then gives the displayed variational characterisation of μ_2 on the complement of the constants.

(Q5) Retraction through the tower. Discretising the Rayleigh quotient on $\mathcal{C}_M$ returns $x^\top \mathcal{L}_{8S} x / x^\top x$, whose minimiser over mean-zero x is the 8S μ_2; the gradient becomes a difference quotient with $O(\ell^2)$ error. Composing

with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \; 0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta^{(12)}_{\text{ind}})]$ vanishes with $\nu \geq 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{12}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: Evolution by conjugation, $A_H(t) = U^\dagger A U$.

Structure: A similarity transformation, so the spectrum of A is constant in time — measurement outcomes are fixed, only their probabilities evolve. Conserved quantities are exactly those commuting with H, which is a statement about shared eigenvectors.

Reduction: None needed.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T2.

Calculus status: Exact linear; carried to the continuum by Lemma D2, and the input already carried a first-order operator, which the transform leaves in place.

13. Ehrenfest's Theorem

Origin (LA edition).

$$\frac{d\langle \mathbf{x} \rangle}{dt} = \frac{\langle \mathbf{p} \rangle}{m}, \qquad \frac{d\langle \mathbf{p} \rangle}{dt} = -\langle \nabla V \rangle$$

Input (8S edition).

$$E^{8S}_{13}(z) = z^\top \mathcal{G}_8 z \Big|_{\text{node } 13}, \qquad \|z_{13}(u) - z_j(u)\|_{\mathcal{G}_8} = \delta^{(8)}_{13j} = \sqrt{(c_{13} - c_j)^\top G_5 (c_{13} - c_j) + (r_{13} - r_j)^2}$$

Calculus image.

$$\mathcal{E}_{13}[\phi] = \tfrac{1}{2} \int_{\mathcal{C}} \|\nabla \phi\|^2_{G_5^{-1}} \, p^2 \, dc, \qquad \delta \mathcal{E}_{13} = 0 \iff \mathcal{L}_\star \phi = 0, \qquad ds^2_{13} = dc^\top (G_5 + \nabla r \, \nabla r^\top) dc + r^2_{13} \, d\Omega^2_3$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T3**, so the calculus image is of class **D3** (Dirichlet energy and the cone metric). The originating equation is first order — it already carries a derivative or an integral. The input already carries a first-order differential or integral operator. On that part the transform acts as the identity — a derivative is already a calculus object — and only the coupling and the gate are taken to the continuum. The regularity demanded of the input is correspondingly one order lower.

(Q2) Image. The displayed relation is the Dirichlet energy and the cone metric of Lemma D3, instantiated at the point $c_{13} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D3 needs **(R3)**, $r \in C^1$ with $r > 0$, for the bundle to be a smooth manifold; at $r = 0$ the fibre collapses and ds^2 degenerates to the base metric. The integration by parts behind Θ-C(1) needs either a Dirichlet or a Neumann boundary condition; on the penteract the latter is the one the kernel limit actually imposes.

(Q4) Derivation. Two derivations meet here. The energy: integrate $\langle \mathcal{L}_\star \phi, \phi \rangle_{p^2}$ by parts, the weight p^2 cancelling against $1/p^2$, to obtain $\frac{1}{2} \int \|\nabla \phi\|^2_{G_5^{-1}} p^2$. The metric: differentiate the embedding $\Phi(c, u) = (c, r(c)u)$, use $u \cdot du = \frac{1}{2} d(\|u\|^2) = 0$ to kill the cross term and $\|u\| = 1$ to normalise the rest, and collect $(\nabla r \cdot dc)^2 = dc^\top \nabla r \nabla r^\top dc$.

(Q5) Retraction through the tower. Replacing ∇ by lattice differences returns the 8S quadratic form $x^\top \mathcal{L}_{8S} x$, and freezing r to a constant collapses ds^2 to the chord $\delta^{(8)}$. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \; 0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta^{(13)}_{\text{ind}})]$ vanishes with $\nu \geq 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{13}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: Expectation values obeying classical-looking equations.

Structure: Exact for quadratic potentials, where $\langle \nabla V \rangle = \nabla V(\langle \mathbf{x} \rangle)$ because the gradient is linear. For anharmonic potentials the two differ — the classical limit is the statement that expectation and nonlinear function commute, which they generally do not.

Reduction: None needed.

Status: Exact linear; classical correspondence exact only for quadratic V.

8S status: Exact linear; classical correspondence exact only for quadratic V; preserved under the transform by Theorem C and Lemma T3.

Calculus status: Exact linear; classical correspondence exact only for quadratic V; carried to the continuum by Lemma D3, and the input already carried a first-order operator, which the transform leaves in place.

14. Probability Current

Origin (LA edition).

$$\mathbf{j} = \frac{\hbar}{2mi}(\psi^* \nabla \psi - \psi \nabla \psi^*)$$

Input (8S edition).

$$E^{8S}_{14}(z) = z^\top \mathcal{G}_8 z\Big|_{\text{node } 14}, \qquad \big\|z_{14}(u) - z_j(u)\big\|_{\mathcal{G}_8} = \delta^{(8)}_{14j} = \sqrt{(c_{14} - c_j)^\top G_5 (c_{14} - c_j) + (r_{14} - r_j)^2}$$

Calculus image.

$$\mathcal{E}_{14}[\phi] = \tfrac{1}{2} \int_{\mathcal{C}} \|\nabla \phi\|^2_{G_5^{-1}}\, p^2\, dc, \qquad \delta \mathcal{E}_{14} = 0 \iff \mathcal{L}_\star \phi = 0, \qquad ds^2_{14} = dc^\top (G_5 + \nabla r\, \nabla r^\top) dc + r^2_{14}\, d\Omega^2_3$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T3**, so the calculus image is of class **D3** (Dirichlet energy and the cone metric). The originating equation is first order — it already carries a derivative or an integral. The input already carries a first-order differential or integral operator. On that part the transform acts as the identity — a derivative is already a calculus object — and only the coupling and the gate are taken to the continuum. The regularity demanded of the input is correspondingly one order lower.

(Q2) Image. The displayed relation is the Dirichlet energy and the cone metric of Lemma D3, instantiated at the point $c_{14} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D3 needs **(R3)**, $r \in C^1$ with $r > 0$, for the bundle to be a smooth manifold; at $r = 0$ the fibre collapses and ds^2 degenerates to the base metric. The integration by parts behind Θ-C(1) needs either a Dirichlet or a Neumann boundary condition; on the penteract the latter is the one the kernel limit actually imposes.

(Q4) Derivation. Two derivations meet here. The energy: integrate $\langle \mathcal{L}_\star \phi, \phi \rangle_{p^2}$ by parts, the weight p^2 cancelling against $1/p^2$, to obtain $\frac{1}{2} \int \|\nabla \phi\|^2_{G_5^{-1}} p^2$. The metric: differentiate the embedding $\Phi(c, u) = (c, r(c)u)$, use $u \cdot du = \frac{1}{2} d(\|u\|^2) = 0$ to kill the cross term and $\|u\| = 1$ to normalise the rest, and collect $(\nabla r \cdot dc)^2 = dc^\top \nabla r \nabla r^\top dc$.

(Q5) Retraction through the tower. Replacing ∇ by lattice differences returns the 8S quadratic form $x^\top \mathcal{L}_{8S} x$, and freezing r to a constant collapses ds^2 to the chord $\delta^{(8)}$. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \; 0 \,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta^{(14)}_{\text{ind}})]$ vanishes with $\nu \geq 0$ and complementary slackness holds. By Θ-A(3) the gradient of the

novelty is $2w_{14}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: A bilinear (sesquilinear) form in the state.

Structure: Together with $|\psi|^2$ it satisfies a continuity equation, which follows from Hermiticity of H — probability conservation is unitarity in local form. The current is quadratic in ψ, so it does not superpose.

Reduction: None needed.

Status: Exact linear (sesquilinear form).

8S status: Exact linear (sesquilinear form); preserved under the transform by Theorem C and Lemma T3.

Calculus status: Exact linear (sesquilinear form); carried to the continuum by Lemma D3, and the input already carried a first-order operator, which the transform leaves in place.

15. Density Matrix

Origin (LA edition).

$$\rho = \sum_i p_i \, |\psi_i\rangle \, \langle \psi_i| \, , \qquad \rho = \rho^\dagger \succeq 0, \ \operatorname{tr} \rho = 1$$

Input (8S edition).

$$\mathcal{L}_{8S} \, \phi_k^{(15)} = \mu_k^{(15)} \phi_k^{(15)}, \qquad \mathcal{L}_{8S} = K_2 L_W + 2K_3 L_{\hat{H}}, \qquad 0 = \mu_1^{(15)} \le \mu_2^{(15)} \le \cdots \le \mu_N^{(15)}$$

Calculus image.

$$-\frac{1}{2p^2} \nabla \cdot (p^2 G_5^{-1} \nabla \phi_k^{(15)}) = \mu_k^{(15)} \phi_k^{(15)}, \qquad \mu_2^{(15)} = \min_{\substack{\phi \neq 0 \\ \int \phi \, p^2 dc = 0}} \frac{\int_{\mathcal{C}} \|\nabla \phi\|^2_{G_5^{-1}} \, p^2 \, dc}{\int_{\mathcal{C}} \phi^2 \, p^2 \, dc}$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T2**, so the calculus image is of class **D2** (Sturm–Liouville problem and Rayleigh quotient). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the Sturm–Liouville problem and Rayleigh quotient of Lemma D2, instantiated at the point $c_{15} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D2 needs **(R1)**, **(R2)** and the standing normalisation **(N)**. Discreteness of the spectrum uses compactness of $\mathcal{C}$ and the Rellich–Kondrachov embedding $H^1 \hookrightarrow L^2$; without compactness the spectrum could be continuous and μ_2 would not be attained. The min–max formula needs self-adjointness on $L^2(p^2 dc)$, which is Θ-C(2).

(Q4) Derivation. Θ-G supplies the operator by the Gaussian-moment expansion; Θ-C(1) converts it into the bilinear form by the divergence theorem, the boundary term vanishing under the Neumann condition the kernel limit imposes. Imposing the normalisation $\int \phi^2 p^2 = 1$ and taking the first variation of $\mathcal{E}$ produces the eigenvalue equation with μ as the Lagrange multiplier; Courant–Fischer then gives the displayed variational characterisation of μ_2 on the complement of the constants.

(Q5) Retraction through the tower. Discretising the Rayleigh quotient on $\mathcal{C}_M$ returns $x^\top \mathcal{L}_{8S} x / x^\top x$, whose minimiser over mean-zero x is the 8S μ_2; the gradient becomes a difference quotient with $O(\ell^2)$ error. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \ 0 \,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta_{\text{ind}}^{(15)})]$ vanishes with $\nu \ge 0$ and complementary slackness holds. By Θ-A(3) the gradient of the

novelty is $2w_{15}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: A Hermitian positive semidefinite unit-trace matrix.

Structure: Pure states are exactly the rank-one ρ ($\operatorname{tr}\rho^2 = 1$); mixed states have higher rank. Purity $\operatorname{tr}\rho^2$ is a rank-like measure, and the eigenvalues of ρ are the mixing probabilities in the optimal decomposition.

Reduction: None needed.

Status: Exact linear. Purity is a rank statement.

8S status: Exact linear. Purity is a rank statement; preserved under the transform by Theorem C and Lemma T2.

Calculus status: Exact linear. Purity is a rank statement; carried to the continuum by Lemma D2, and the input was algebraic, so the differential content is entirely supplied by the transform.

Also in: Statistical Mechanics.

16. Von Neumann Equation

Origin (LA edition).

$$i\hbar\dot{\rho} = [H, \rho]$$

Input (8S edition).

$$\mathcal{L}_{8S}\,\phi_k^{(16)} = \mu_k^{(16)}\phi_k^{(16)}, \qquad \mathcal{L}_{8S} = K_2 L_W + 2K_3 L_{\hat{H}}, \qquad 0 = \mu_1^{(16)} \le \mu_2^{(16)} \le \cdots \le \mu_N^{(16)}$$

Calculus image.

$$-\frac{1}{2p^2}\nabla\cdot(p^2 G_5^{-1}\nabla\phi_k^{(16)}) = \mu_k^{(16)}\phi_k^{(16)}, \qquad \mu_2^{(16)} = \min_{\substack{\phi\neq 0 \\ \int \phi\, p^2\, dc = 0}} \frac{\int_{\mathcal{C}} \|\nabla\phi\|^2_{G_5^{-1}}\, p^2\, dc}{\int_{\mathcal{C}} \phi^2\, p^2\, dc}$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T2**, so the calculus image is of class **D2** (Sturm–Liouville problem and Rayleigh quotient). The originating equation is first order — it already carries a derivative or an integral. The input already carries a first-order differential or integral operator. On that part the transform acts as the identity — a derivative is already a calculus object — and only the coupling and the gate are taken to the continuum. The regularity demanded of the input is correspondingly one order lower.

(Q2) Image. The displayed relation is the Sturm–Liouville problem and Rayleigh quotient of Lemma D2, instantiated at the point $c_{16} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D2 needs **(R1)**, **(R2)** and the standing normalisation **(N)**. Discreteness of the spectrum uses compactness of $\mathcal{C}$ and the Rellich–Kondrachov embedding $H^1 \hookrightarrow L^2$; without compactness the spectrum could be continuous and μ_2 would not be attained. The min–max formula needs self-adjointness on $L^2(p^2 dc)$, which is Θ-C(2).

(Q4) Derivation. Θ-G supplies the operator by the Gaussian-moment expansion; Θ-C(1) converts it into the bilinear form by the divergence theorem, the boundary term vanishing under the Neumann condition the kernel limit imposes. Imposing the normalisation $\int \phi^2 p^2 = 1$ and taking the first variation of $\mathcal{E}$ produces the eigenvalue equation with μ as the Lagrange multiplier; Courant–Fischer then gives the displayed variational characterisation of μ_2 on the complement of the constants.

(Q5) Retraction through the tower. Discretising the Rayleigh quotient on $\mathcal{C}_M$ returns $x^\top \mathcal{L}_{8S} x / x^\top x$, whose minimiser over mean-zero x is the 8S μ_2; the gradient becomes a difference quotient with $O(\ell^2)$ error. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \;\, 0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta_{\text{ind}}^{(16)})]$ vanishes with $\nu \geq 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{16}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: Evolution by commutator — the adjoint action.

Structure: $\rho(t) = U\rho U^\dagger$, a unitary congruence, so the eigenvalues of ρ are constant. Von Neumann entropy is therefore conserved under closed-system evolution — irreversibility requires opening the system.

Reduction: None needed.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T2.

Calculus status: Exact linear; carried to the continuum by Lemma D2, and the input already carried a first-order operator, which the transform leaves in place.

17. Feynman Path Integral

Origin (LA edition).

$$\langle x_f \mid U \mid x_i \rangle = \int \mathcal{D}[x]\, e^{iS[x]/\hbar}$$

Input (8S edition).

$$\operatorname{rank}(C_5) = \operatorname{rank}(C_4) + \mathbf{1}[\eta_{\text{ind}}^{(17)} > 0], \qquad \dim\ker \mathcal{L}_{8S} = \#\operatorname{comp}(W + \hat{H})$$

Calculus image.

$$\dim\ker \mathcal{L}_\star = \dim H^0(\mathcal{C}) = \#\{\text{components}\}, \qquad \operatorname{ind} \mathcal{L}_\star = 0, \qquad \det \operatorname{Gr}(C_4, e_{17}) \neq 0 \iff \text{rank increases}$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T5**, so the calculus image is of class **D5** (kernel dimension, index, and H^0). The originating equation is first order — it already carries a derivative or an integral. The input already carries a first-order differential or integral operator. On that part the transform acts as the identity — a derivative is already a calculus object — and only the coupling and the gate are taken to the continuum. The regularity demanded of the input is correspondingly one order lower.

(Q2) Image. The displayed relation is the kernel dimension, index, and H^0 of Lemma D5, instantiated at the point $c_{17} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D5 needs $\mathcal{L}_\star$ self-adjoint with compact resolvent, which is Θ-G under **(R1)**–**(R2)**; the identification of ker with locally constant functions needs $G_5 \succ 0$ and $p > 0$, both standing. The index vanishes because the operator is self-adjoint, so coker ≅ ker.

(Q4) Derivation. Set $\mathcal{E}[\phi] = 0$ in Θ-C(1). Since $G_5^{-1} \succ 0$ and $p > 0$, this forces $\nabla\phi = 0$ almost everywhere, hence ϕ locally constant, hence $\ker \mathcal{L}_\star$ spanned by the indicator functions of the connected components — which is precisely H^0. Self-adjointness gives coker ≅ ker and therefore index zero.

(Q5) Retraction through the tower. On the lattice $\dim\ker \mathcal{L}_\star$ becomes $\dim\ker \mathcal{L}_{8S}$, the component count of the coupling graph, and the Gram condition becomes the 8S rank increment. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \; 0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta_{\text{ind}}^{(17)})]$ vanishes with $\nu \geq 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{17}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: A sum over paths reproducing a matrix element.

Structure: Equivalent to the matrix exponential: Trotter splitting $e^{-iHt/\hbar} = \lim(e^{-iVt/N}e^{-iTt/N})^N$ turns the propagator into a product of matrices, and taking the continuum limit gives the path integral. It is a factorisation of a matrix exponential.

Reduction: The path integral is a Trotter product of matrices.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T5.

Calculus status: Exact linear; carried to the continuum by Lemma D5, and the input already carried a first-order operator, which the transform leaves in place.

Exactly Solvable Systems

18. Infinite Square Well

Origin (LA edition).

$$E_n = \frac{n^2\pi^2\hbar^2}{2mL^2}, \qquad \psi_n = \sqrt{\tfrac{2}{L}}\sin\frac{n\pi x}{L}$$

Input (8S edition).

$$\mathcal{L}_{8S}\,\phi_k^{(18)} = \mu_k^{(18)}\phi_k^{(18)}, \qquad \mathcal{L}_{8S} = K_2 L_W + 2K_3 L_{\hat{H}}, \qquad 0 = \mu_1^{(18)} \le \mu_2^{(18)} \le \cdots \le \mu_N^{(18)}$$

Calculus image.

$$-\frac{1}{2p^2}\nabla\cdot(p^2 G_5^{-1}\nabla\phi_k^{(18)}) = \mu_k^{(18)}\phi_k^{(18)}, \qquad \mu_2^{(18)} = \min_{\substack{\phi\neq 0 \\ \int \phi\, p^2 dc = 0}} \frac{\int_{\mathcal{C}} \|\nabla\phi\|^2_{G_5^{-1}}\, p^2\, dc}{\int_{\mathcal{C}} \phi^2\, p^2\, dc}$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T2**, so the calculus image is of class **D2** (Sturm–Liouville problem and Rayleigh quotient). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the Sturm–Liouville problem and Rayleigh quotient of Lemma D2, instantiated at the point $c_{18} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D2 needs **(R1)**, **(R2)** and the standing normalisation **(N)**. Discreteness of the spectrum uses compactness of $\mathcal{C}$ and the Rellich–Kondrachov embedding $H^1 \hookrightarrow L^2$; without compactness the spectrum could be continuous and μ_2 would not be attained. The min–max formula needs self-adjointness on $L^2(p^2dc)$, which is Θ-C(2).

(Q4) Derivation. Θ-G supplies the operator by the Gaussian-moment expansion; Θ-C(1) converts it into the bilinear form by the divergence theorem, the boundary term vanishing under the Neumann condition the kernel limit imposes. Imposing the normalisation $\int \phi^2 p^2 = 1$ and taking the first variation of $\mathcal{E}$ produces the eigenvalue equation with μ as the Lagrange multiplier; Courant–Fischer then gives the displayed variational characterisation of μ_2 on the complement of the constants.

(Q5) Retraction through the tower. Discretising the Rayleigh quotient on $\mathcal{C}_M$ returns $x^\top \mathcal{L}_{8S} x / x^\top x$, whose minimiser over mean-zero x is the 8S μ_2; the gradient becomes a difference quotient with $O(\ell^2)$ error. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\,I_3\ 0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta_{\text{ind}}^{(18)})]$ vanishes with $\nu \ge 0$ and complementary slackness holds. By Θ-A(3) the gradient of the

novelty is $2w_{18}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: Eigenvalues of the Dirichlet Laplacian on an interval.

Structure: Exactly the vibrating string problem with a different constant — the same operator, the same eigenvectors. Quantisation is discreteness of the spectrum, which comes from the boundary conditions, not from anything quantum.

Reduction: None needed.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T2.

Calculus status: Exact linear; carried to the continuum by Lemma D2, and the input was algebraic, so the differential content is entirely supplied by the transform.

19. Finite Square Well Bound States

Origin (LA edition).

$$\tan(k_1 a) = \frac{k_2}{k_1} \text{ (even states)}$$

Input (8S edition).

$$\mathcal{L}_{8S}\,\phi_k^{(19)} = \mu_k^{(19)}\phi_k^{(19)}, \qquad \mathcal{L}_{8S} = K_2 L_W + 2K_3 L_{\hat{H}}, \qquad 0 = \mu_1^{(19)} \le \mu_2^{(19)} \le \cdots \le \mu_N^{(19)}$$

Calculus image.

$$-\frac{1}{2p^2}\nabla\cdot(p^2 G_5^{-1}\nabla\phi_k^{(19)}) = \mu_k^{(19)}\phi_k^{(19)}, \qquad \mu_2^{(19)} = \min_{\substack{\phi\neq 0 \\ \int \phi\, p^2 dc = 0}} \frac{\int_{\mathcal{C}} \|\nabla\phi\|^2_{G_5^{-1}}\, p^2\, dc}{\int_{\mathcal{C}} \phi^2\, p^2\, dc}$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T2**, so the calculus image is of class **D2** (Sturm–Liouville problem and Rayleigh quotient). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the Sturm–Liouville problem and Rayleigh quotient of Lemma D2, instantiated at the point $c_{19} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D2 needs **(R1)**, **(R2)** and the standing normalisation **(N)**. Discreteness of the spectrum uses compactness of $\mathcal{C}$ and the Rellich–Kondrachov embedding $H^1 \hookrightarrow L^2$; without compactness the spectrum could be continuous and μ_2 would not be attained. The min–max formula needs self-adjointness on $L^2(p^2 dc)$, which is Θ-C(2).

(Q4) Derivation. Θ-G supplies the operator by the Gaussian-moment expansion; Θ-C(1) converts it into the bilinear form by the divergence theorem, the boundary term vanishing under the Neumann condition the kernel limit imposes. Imposing the normalisation $\int \phi^2 p^2 = 1$ and taking the first variation of $\mathcal{E}$ produces the eigenvalue equation with μ as the Lagrange multiplier; Courant–Fischer then gives the displayed variational characterisation of μ_2 on the complement of the constants.

(Q5) Retraction through the tower. Discretising the Rayleigh quotient on $\mathcal{C}_M$ returns $x^\top \mathcal{L}_{8S} x / x^\top x$, whose minimiser over mean-zero x is the 8S μ_2; the gradient becomes a difference quotient with $O(\ell^2)$ error. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \;\, 0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta_{\text{ind}}^{(19)})]$ vanishes with $\nu \ge 0$ and complementary slackness holds. By Θ-A(3) the gradient of the

novelty is $2w_{19}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: A transcendental eigenvalue condition.

Structure: Matching conditions at the boundaries give a 2×2 transfer matrix; bound states are where its determinant condition is met. The number of bound states is $\lceil 2a\sqrt{2mV_0}/\pi\hbar \rceil$ — an eigenvalue count, always at least one in one dimension.

Reduction: None needed.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T2.

Calculus status: Exact linear; carried to the continuum by Lemma D2, and the input was algebraic, so the differential content is entirely supplied by the transform.

20. Quantum Harmonic Oscillator

Origin (LA edition).

$$E_n = \hbar\omega\left(n + \tfrac{1}{2}\right)$$

Input (8S edition).

$$\mathcal{L}_{8S}\,\phi_k^{(20)} = \mu_k^{(20)}\phi_k^{(20)}, \qquad \mathcal{L}_{8S} = K_2 L_W + 2K_3 L_{\hat{H}}, \qquad 0 = \mu_1^{(20)} \le \mu_2^{(20)} \le \cdots \le \mu_N^{(20)}$$

Calculus image.

$$-\frac{1}{2p^2}\nabla\cdot(p^2 G_5^{-1}\nabla\phi_k^{(20)}) = \mu_k^{(20)}\phi_k^{(20)}, \qquad \mu_2^{(20)} = \min_{\substack{\phi\neq 0 \\ \int \phi\, p^2\, dc = 0}} \frac{\int_{\mathcal{C}} \|\nabla\phi\|^2_{G_5^{-1}}\, p^2\, dc}{\int_{\mathcal{C}} \phi^2\, p^2\, dc}$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T2**, so the calculus image is of class **D2** (Sturm–Liouville problem and Rayleigh quotient). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the Sturm–Liouville problem and Rayleigh quotient of Lemma D2, instantiated at the point $c_{20} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D2 needs **(R1)**, **(R2)** and the standing normalisation **(N)**. Discreteness of the spectrum uses compactness of $\mathcal{C}$ and the Rellich–Kondrachov embedding $H^1 \hookrightarrow L^2$; without compactness the spectrum could be continuous and μ_2 would not be attained. The min–max formula needs self-adjointness on $L^2(p^2 dc)$, which is Θ-C(2).

(Q4) Derivation. Θ-G supplies the operator by the Gaussian-moment expansion; Θ-C(1) converts it into the bilinear form by the divergence theorem, the boundary term vanishing under the Neumann condition the kernel limit imposes. Imposing the normalisation $\int \phi^2 p^2 = 1$ and taking the first variation of $\mathcal{E}$ produces the eigenvalue equation with μ as the Lagrange multiplier; Courant–Fischer then gives the displayed variational characterisation of μ_2 on the complement of the constants.

(Q5) Retraction through the tower. Discretising the Rayleigh quotient on $\mathcal{C}_M$ returns $x^\top \mathcal{L}_{8S} x / x^\top x$, whose minimiser over mean-zero x is the 8S μ_2; the gradient becomes a difference quotient with $O(\ell^2)$ error. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \;\, 0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta_{\text{ind}}^{(20)})]$ vanishes with $\nu \ge 0$ and complementary slackness holds. By Θ-A(3) the gradient of the

novelty is $2w_{20}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: An arithmetic spectrum — evenly spaced eigenvalues.

Structure: The equal spacing is what makes the oscillator special: it permits the ladder construction, makes coherent states possible, and gives the geometric partition function. In the number basis H is diagonal and a is the subdiagonal shift matrix.

Reduction: None needed.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T2.

Calculus status: Exact linear; carried to the continuum by Lemma D2, and the input was algebraic, so the differential content is entirely supplied by the transform.

21. Ladder Operators

Origin (LA edition).

$$a = \sqrt{\frac{m\omega}{2\hbar}}\left(\hat{x} + \frac{i\hat{p}}{m\omega}\right), \qquad [a, a^\dagger] = 1$$

Input (8S edition).

$$\mathcal{L}_{8S}\,\phi_k^{(21)} = \mu_k^{(21)}\phi_k^{(21)}, \qquad \mathcal{L}_{8S} = K_2 L_W + 2K_3 L_{\hat{H}}, \qquad 0 = \mu_1^{(21)} \le \mu_2^{(21)} \le \cdots \le \mu_N^{(21)}$$

Calculus image.

$$-\frac{1}{2p^2}\nabla\cdot(p^2 G_5^{-1}\nabla\phi_k^{(21)}) = \mu_k^{(21)}\phi_k^{(21)}, \qquad \mu_2^{(21)} = \min_{\substack{\phi\neq 0 \\ \int \phi\, p^2 dc = 0}} \frac{\int_{\mathcal{C}} \|\nabla\phi\|^2_{G_5^{-1}}\, p^2\, dc}{\int_{\mathcal{C}} \phi^2\, p^2\, dc}$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T2**, so the calculus image is of class **D2** (Sturm–Liouville problem and Rayleigh quotient). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the Sturm–Liouville problem and Rayleigh quotient of Lemma D2, instantiated at the point $c_{21} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D2 needs **(R1)**, **(R2)** and the standing normalisation **(N)**. Discreteness of the spectrum uses compactness of $\mathcal{C}$ and the Rellich–Kondrachov embedding $H^1 \hookrightarrow L^2$; without compactness the spectrum could be continuous and μ_2 would not be attained. The min–max formula needs self-adjointness on $L^2(p^2 dc)$, which is Θ-C(2).

(Q4) Derivation. Θ-G supplies the operator by the Gaussian-moment expansion; Θ-C(1) converts it into the bilinear form by the divergence theorem, the boundary term vanishing under the Neumann condition the kernel limit imposes. Imposing the normalisation $\int \phi^2 p^2 = 1$ and taking the first variation of $\mathcal{E}$ produces the eigenvalue equation with μ as the Lagrange multiplier; Courant–Fischer then gives the displayed variational characterisation of μ_2 on the complement of the constants.

(Q5) Retraction through the tower. Discretising the Rayleigh quotient on $\mathcal{C}_M$ returns $x^\top \mathcal{L}_{8S} x / x^\top x$, whose minimiser over mean-zero x is the 8S μ_2; the gradient becomes a difference quotient with $O(\ell^2)$ error. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \;\, 0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta_{\text{ind}}^{(21)})]$ vanishes with $\nu \ge 0$ and complementary slackness holds. By Θ-A(3) the gradient of the

novelty is $2w_{21}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: Explicit shift matrices in the number basis.

Structure: a is the subdiagonal with entries $\sqrt{n}$ and $a^\dagger$ its transpose; $H = \hbar\omega(a^\dagger a + \frac{1}{2})$ is diagonal. The whole spectrum follows from one commutator and the existence of a lowest eigenvector — a purely algebraic construction requiring no differential equations.

Reduction: None needed.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T2.

Calculus status: Exact linear; carried to the continuum by Lemma D2, and the input was algebraic, so the differential content is entirely supplied by the transform.

22. Coherent State

Origin (LA edition).

$$a\,|\alpha\rangle = \alpha\,|\alpha\rangle\,, \qquad |\alpha\rangle = e^{-|\alpha|^2/2}\sum_n \frac{\alpha^n}{\sqrt{n!}}\,|n\rangle$$

Input (8S edition).

$$\mathcal{L}_{8S}\,\phi_k^{(22)} = \mu_k^{(22)}\phi_k^{(22)}, \qquad \mathcal{L}_{8S} = K_2 L_W + 2K_3 L_{\hat{H}}, \qquad 0 = \mu_1^{(22)} \le \mu_2^{(22)} \le \cdots \le \mu_N^{(22)}$$

Calculus image.

$$-\frac{1}{2p^2}\nabla\cdot(p^2 G_5^{-1}\nabla\phi_k^{(22)}) = \mu_k^{(22)}\phi_k^{(22)}, \qquad \mu_2^{(22)} = \min_{\substack{\phi\neq 0\\ \int \phi\, p^2\, dc = 0}} \frac{\int_{\mathcal{C}} \|\nabla\phi\|^2_{G_5^{-1}}\, p^2\, dc}{\int_{\mathcal{C}} \phi^2\, p^2\, dc}$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T2**, so the calculus image is of class **D2** (Sturm–Liouville problem and Rayleigh quotient). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the Sturm–Liouville problem and Rayleigh quotient of Lemma D2, instantiated at the point $c_{22} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D2 needs **(R1)**, **(R2)** and the standing normalisation **(N)**. Discreteness of the spectrum uses compactness of $\mathcal{C}$ and the Rellich–Kondrachov embedding $H^1 \hookrightarrow L^2$; without compactness the spectrum could be continuous and μ_2 would not be attained. The min–max formula needs self-adjointness on $L^2(p^2 dc)$, which is Θ-C(2).

(Q4) Derivation. Θ-G supplies the operator by the Gaussian-moment expansion; Θ-C(1) converts it into the bilinear form by the divergence theorem, the boundary term vanishing under the Neumann condition the kernel limit imposes. Imposing the normalisation $\int \phi^2 p^2 = 1$ and taking the first variation of $\mathcal{E}$ produces the eigenvalue equation with μ as the Lagrange multiplier; Courant–Fischer then gives the displayed variational characterisation of μ_2 on the complement of the constants.

(Q5) Retraction through the tower. Discretising the Rayleigh quotient on $\mathcal{C}_M$ returns $x^\top \mathcal{L}_{8S} x / x^\top x$, whose minimiser over mean-zero x is the 8S μ_2; the gradient becomes a difference quotient with $O(\ell^2)$ error. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\,I_3\ 0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta_{\text{ind}}^{(22)})]$ vanishes with $\nu \ge 0$ and complementary slackness holds. By Θ-A(3) the gradient of the

novelty is $2w_{22}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: An eigenvector of the non-Hermitian operator a.

Structure: Since a is not normal, its eigenvectors are not orthogonal — coherent states form an overcomplete, non-orthogonal frame. They saturate the uncertainty relation and are the closest quantum analogue of a classical state.

Reduction: None needed.

Status: Exact linear (non-Hermitian eigenproblem).

8S status: Exact linear (non-Hermitian eigenproblem); preserved under the transform by Theorem C and Lemma T2.

Calculus status: Exact linear (non-Hermitian eigenproblem); carried to the continuum by Lemma D2, and the input was algebraic, so the differential content is entirely supplied by the transform.

23. Quantum Tunnelling Transmission

Origin (LA edition).

$$T \approx e^{-2\kappa L}, \qquad \kappa = \frac{\sqrt{2m(V_0 - E)}}{\hbar}$$

Input (8S edition).

$$\mathcal{L}_{8S}\,\phi_k^{(23)} = \mu_k^{(23)}\phi_k^{(23)}, \qquad \mathcal{L}_{8S} = K_2 L_W + 2K_3 L_{\hat{H}}, \qquad 0 = \mu_1^{(23)} \le \mu_2^{(23)} \le \cdots \le \mu_N^{(23)}$$

Calculus image.

$$-\frac{1}{2p^2}\nabla\cdot(p^2 G_5^{-1}\nabla\phi_k^{(23)}) = \mu_k^{(23)}\phi_k^{(23)}, \qquad \mu_2^{(23)} = \min_{\substack{\phi\neq 0 \\ \int \phi\, p^2\, dc = 0}} \frac{\int_{\mathcal{C}} \|\nabla\phi\|^2_{G_5^{-1}}\, p^2\, dc}{\int_{\mathcal{C}} \phi^2\, p^2\, dc}$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T2**, so the calculus image is of class **D2** (Sturm–Liouville problem and Rayleigh quotient). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the Sturm–Liouville problem and Rayleigh quotient of Lemma D2, instantiated at the point $c_{23} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D2 needs **(R1)**, **(R2)** and the standing normalisation **(N)**. Discreteness of the spectrum uses compactness of $\mathcal{C}$ and the Rellich–Kondrachov embedding $H^1 \hookrightarrow L^2$; without compactness the spectrum could be continuous and μ_2 would not be attained. The min–max formula needs self-adjointness on $L^2(p^2 dc)$, which is Θ-C(2).

(Q4) Derivation. Θ-G supplies the operator by the Gaussian-moment expansion; Θ-C(1) converts it into the bilinear form by the divergence theorem, the boundary term vanishing under the Neumann condition the kernel limit imposes. Imposing the normalisation $\int \phi^2 p^2 = 1$ and taking the first variation of $\mathcal{E}$ produces the eigenvalue equation with μ as the Lagrange multiplier; Courant–Fischer then gives the displayed variational characterisation of μ_2 on the complement of the constants.

(Q5) Retraction through the tower. Discretising the Rayleigh quotient on $\mathcal{C}_M$ returns $x^\top \mathcal{L}_{8S} x / x^\top x$, whose minimiser over mean-zero x is the 8S μ_2; the gradient becomes a difference quotient with $O(\ell^2)$ error. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \;\, 0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta_{\text{ind}}^{(23)})]$ vanishes with $\nu \geq 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{23}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: An entry of the barrier transfer matrix.

Structure: The barrier is a 2×2 transfer matrix whose off-diagonal entry gives transmission. Inside the barrier the wavevector is imaginary, so the solution decays — evanescence is exactly an eigenvalue leaving the real axis.

Reduction: None needed.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T2.

Calculus status: Exact linear; carried to the continuum by Lemma D2, and the input was algebraic, so the differential content is entirely supplied by the transform.

24. WKB Tunnelling Integral

Origin (LA edition).

$$T \approx \exp\Big(-\frac{2}{\hbar}\int \sqrt{2m(V-E)}\,dx\Big)$$

Input (8S edition).

$$\mathcal{L}_{8S}\,\phi_k^{(24)} = \mu_k^{(24)}\phi_k^{(24)}, \qquad \mathcal{L}_{8S} = K_2 L_W + 2K_3 L_{\hat{H}}, \qquad 0 = \mu_1^{(24)} \leq \mu_2^{(24)} \leq \cdots \leq \mu_N^{(24)}$$

Calculus image.

$$-\frac{1}{2p^2}\nabla\cdot(p^2 G_5^{-1}\nabla\phi_k^{(24)}) = \mu_k^{(24)}\phi_k^{(24)}, \qquad \mu_2^{(24)} = \min_{\substack{\phi\neq 0 \\ \int \phi\, p^2 dc = 0}} \frac{\int_{\mathcal{C}} \|\nabla\phi\|^2_{G_5^{-1}}\, p^2\, dc}{\int_{\mathcal{C}} \phi^2\, p^2\, dc}$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T2**, so the calculus image is of class **D2** (Sturm–Liouville problem and Rayleigh quotient). The originating equation is first order — it already carries a derivative or an integral. The input already carries a first-order differential or integral operator. On that part the transform acts as the identity — a derivative is already a calculus object — and only the coupling and the gate are taken to the continuum. The regularity demanded of the input is correspondingly one order lower.

(Q2) Image. The displayed relation is the Sturm–Liouville problem and Rayleigh quotient of Lemma D2, instantiated at the point $c_{24} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D2 needs **(R1)**, **(R2)** and the standing normalisation **(N)**. Discreteness of the spectrum uses compactness of $\mathcal{C}$ and the Rellich–Kondrachov embedding $H^1 \hookrightarrow L^2$; without compactness the spectrum could be continuous and μ_2 would not be attained. The min–max formula needs self-adjointness on $L^2(p^2 dc)$, which is Θ-C(2).

(Q4) Derivation. Θ-G supplies the operator by the Gaussian-moment expansion; Θ-C(1) converts it into the bilinear form by the divergence theorem, the boundary term vanishing under the Neumann condition the kernel limit imposes. Imposing the normalisation $\int \phi^2 p^2 = 1$ and taking the first variation of $\mathcal{E}$ produces the eigenvalue equation with μ as the Lagrange multiplier; Courant–Fischer then gives the displayed variational characterisation of μ_2 on the complement of the constants.

(Q5) Retraction through the tower. Discretising the Rayleigh quotient on $\mathcal{C}_M$ returns $x^\top \mathcal{L}_{8S} x / x^\top x$, whose minimiser over mean-zero x is the 8S μ_2; the gradient becomes a difference quotient with $O(\ell^2)$ error. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\,I_3\ 0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta_{\text{ind}}^{(24)})]$ vanishes with $\nu \geq 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{24}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: The adiabatic approximation to the transfer matrix.

Structure: Assumes the local eigenvalue varies slowly, so the transfer matrix is a product of locally diagonal factors. It fails at turning points where the eigenvalues collide, requiring Airy connection formulae.

Reduction: WKB is an adiabatic product of local matrices.

Status: Linearised (adiabatic approximation).

8S status: Linearised (adiabatic approximation); the 8S image inherits the same approximation status (Theorem C, Lemma T2).

Calculus status: Linearised (adiabatic approximation); carried to the continuum by Lemma D2, and the input already carried a first-order operator, which the transform leaves in place.

25. Hydrogen Atom Energy Levels

Origin (LA edition).

$$E_n = -\frac{13.6 \text{ eV}}{n^2}$$

Input (8S edition).

$$\mathcal{L}_{8S}\,\phi_k^{(25)} = \mu_k^{(25)}\phi_k^{(25)}, \qquad \mathcal{L}_{8S} = K_2 L_W + 2K_3 L_{\hat{H}}, \qquad 0 = \mu_1^{(25)} \leq \mu_2^{(25)} \leq \cdots \leq \mu_N^{(25)}$$

Calculus image.

$$-\frac{1}{2p^2}\nabla\cdot(p^2 G_5^{-1}\nabla\phi_k^{(25)}) = \mu_k^{(25)}\phi_k^{(25)}, \qquad \mu_2^{(25)} = \min_{\substack{\phi\neq 0 \\ \int \phi\, p^2 dc = 0}} \frac{\int_{\mathcal{C}} \|\nabla\phi\|^2_{G_5^{-1}}\, p^2\, dc}{\int_{\mathcal{C}} \phi^2\, p^2\, dc}$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T2**, so the calculus image is of class **D2** (Sturm–Liouville problem and Rayleigh quotient). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the Sturm–Liouville problem and Rayleigh quotient of Lemma D2, instantiated at the point $c_{25} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D2 needs **(R1)**, **(R2)** and the standing normalisation **(N)**. Discreteness of the spectrum uses compactness of $\mathcal{C}$ and the Rellich–Kondrachov embedding $H^1 \hookrightarrow L^2$; without compactness the spectrum could be continuous and μ_2 would not be attained. The min–max formula needs self-adjointness on $L^2(p^2 dc)$, which is Θ-C(2).

(Q4) Derivation. Θ-G supplies the operator by the Gaussian-moment expansion; Θ-C(1) converts it into the bilinear form by the divergence theorem, the boundary term vanishing under the Neumann condition the kernel limit imposes. Imposing the normalisation $\int \phi^2 p^2 = 1$ and taking the first variation of $\mathcal{E}$ produces the eigenvalue equation with μ as the Lagrange multiplier; Courant–Fischer then gives the displayed variational characterisation of μ_2 on the complement of the constants.

(Q5) Retraction through the tower. Discretising the Rayleigh quotient on $\mathcal{C}_M$ returns $x^\top \mathcal{L}_{8S} x / x^\top x$, whose minimiser over mean-zero x is the 8S μ_2; the gradient becomes a difference quotient with $O(\ell^2)$ error. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \; 0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta_{\text{ind}}^{(25)})]$ vanishes with $\nu \geq 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{25}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: The Coulomb Hamiltonian's discrete spectrum.

Structure: The n^2-fold degeneracy is larger than rotational symmetry alone predicts — the extra degeneracy comes from the hidden $SO(4)$ symmetry generated by the Runge–Lenz vector. Degeneracy is always a symmetry, and here the symmetry is bigger than it looks.

Reduction: None needed.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T2.

Calculus status: Exact linear; carried to the continuum by Lemma D2, and the input was algebraic, so the differential content is entirely supplied by the transform.

26. Bohr Radius

Origin (LA edition).

$$a_0 = \frac{4\pi\epsilon_0\hbar^2}{me^2} = 0.529 \text{ Å}$$

Input (8S edition).

$$\mathcal{L}_{8S}\,\phi_k^{(26)} = \mu_k^{(26)}\phi_k^{(26)}, \qquad \mathcal{L}_{8S} = K_2 L_W + 2K_3 L_{\hat{H}}, \qquad 0 = \mu_1^{(26)} \leq \mu_2^{(26)} \leq \cdots \leq \mu_N^{(26)}$$

Calculus image.

$$-\frac{1}{2p^2}\nabla\cdot(p^2 G_5^{-1}\nabla\phi_k^{(26)}) = \mu_k^{(26)}\phi_k^{(26)}, \qquad \mu_2^{(26)} = \min_{\substack{\phi\neq 0 \\ \int \phi\, p^2 dc = 0}} \frac{\int_{\mathcal{C}} \|\nabla\phi\|^2_{G_5^{-1}}\, p^2\, dc}{\int_{\mathcal{C}} \phi^2\, p^2\, dc}$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T2**, so the calculus image is of class **D2** (Sturm–Liouville problem and Rayleigh quotient). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the Sturm–Liouville problem and Rayleigh quotient of Lemma D2, instantiated at the point $c_{26} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D2 needs **(R1)**, **(R2)** and the standing normalisation **(N)**. Discreteness of the spectrum uses compactness of $\mathcal{C}$ and the Rellich–Kondrachov embedding $H^1 \hookrightarrow L^2$; without compactness the spectrum could be continuous and μ_2 would not be attained. The min–max formula needs self-adjointness on $L^2(p^2 dc)$, which is Θ-C(2).

(Q4) Derivation. Θ-G supplies the operator by the Gaussian-moment expansion; Θ-C(1) converts it into the bilinear form by the divergence theorem, the boundary term vanishing under the Neumann condition the kernel limit imposes. Imposing the normalisation $\int \phi^2 p^2 = 1$ and taking the first variation of $\mathcal{E}$ produces the eigenvalue equation with μ as the Lagrange multiplier; Courant–Fischer then gives the displayed variational characterisation of μ_2 on the complement of the constants.

(Q5) Retraction through the tower. Discretising the Rayleigh quotient on $\mathcal{C}_M$ returns $x^\top \mathcal{L}_{8S} x / x^\top x$, whose minimiser over mean-zero x is the 8S μ_2; the gradient becomes a difference quotient with $O(\ell^2)$ error. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \; 0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta_{\text{ind}}^{(26)})]$ vanishes with $\nu \geq 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{26}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: The length scale from the eigenvalue problem.

Structure: Fixed by dimensional analysis of the Hamiltonian's constants — a null-space computation on the dimensional matrix. It sets the scale of every atomic wavefunction.

Reduction: The scale comes from a dimensional null space.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T2.

Calculus status: Exact linear; carried to the continuum by Lemma D2, and the input was algebraic, so the differential content is entirely supplied by the transform.

27. Hydrogen Wavefunctions

Origin (LA edition).

$$\psi_{n\ell m} = R_{n\ell}(r) Y_\ell^m(\theta, \phi)$$

Input (8S edition).

$$\mathcal{L}_{8S}\, \phi_k^{(27)} = \mu_k^{(27)} \phi_k^{(27)}, \qquad \mathcal{L}_{8S} = K_2 L_W + 2K_3 L_{\hat{H}}, \qquad 0 = \mu_1^{(27)} \leq \mu_2^{(27)} \leq \cdots \leq \mu_N^{(27)}$$

Calculus image.

$$-\frac{1}{2p^2} \nabla \cdot (p^2 G_5^{-1} \nabla \phi_k^{(27)}) = \mu_k^{(27)} \phi_k^{(27)}, \qquad \mu_2^{(27)} = \min_{\substack{\phi \neq 0 \\ \int \phi\, p^2 dc = 0}} \frac{\int_{\mathcal{C}} \|\nabla \phi\|^2_{G_5^{-1}}\, p^2\, dc}{\int_{\mathcal{C}} \phi^2\, p^2\, dc}$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T2**, so the calculus image is of class **D2** (Sturm–Liouville problem and Rayleigh quotient). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the Sturm–Liouville problem and Rayleigh quotient of Lemma D2, instantiated at the point $c_{27} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D2 needs **(R1)**, **(R2)** and the standing normalisation **(N)**. Discreteness of the spectrum uses compactness of $\mathcal{C}$ and the Rellich–Kondrachov embedding $H^1 \hookrightarrow L^2$; without compactness the spectrum could be continuous and μ_2 would not be attained. The min–max formula needs self-adjointness on $L^2(p^2 dc)$, which is Θ-C(2).

(Q4) Derivation. Θ-G supplies the operator by the Gaussian-moment expansion; Θ-C(1) converts it into the bilinear form by the divergence theorem, the boundary term vanishing under the Neumann condition the kernel limit imposes. Imposing the normalisation $\int \phi^2 p^2 = 1$ and taking the first variation of $\mathcal{E}$ produces the eigenvalue equation with μ as the Lagrange multiplier; Courant–Fischer then gives the displayed variational characterisation of μ_2 on the complement of the constants.

(Q5) Retraction through the tower. Discretising the Rayleigh quotient on $\mathcal{C}_M$ returns $x^\top \mathcal{L}_{8S} x / x^\top x$, whose minimiser over mean-zero x is the 8S μ_2; the gradient becomes a difference quotient with $O(\ell^2)$ error. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \;\, 0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta_{\text{ind}}^{(27)})]$ vanishes with $\nu \geq 0$ and complementary slackness holds. By Θ-A(3) the gradient of the

novelty is $2w_{27}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: Eigenvectors, factorised by separation of variables.

Structure: Separability reflects the Kronecker sum structure of the Hamiltonian in spherical coordinates, so eigenvectors are products. The angular factors are $SO(3)$ irreducible representation basis vectors; the radial factors are Laguerre polynomials.

Reduction: None needed.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T2.

Calculus status: Exact linear; carried to the continuum by Lemma D2, and the input was algebraic, so the differential content is entirely supplied by the transform.

28. Rydberg Formula

Origin (LA edition).

$$\frac{1}{\lambda} = R\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)$$

Input (8S edition).

$$\mathcal{L}_{8S}\,\phi_k^{(28)} = \mu_k^{(28)}\phi_k^{(28)}, \qquad \mathcal{L}_{8S} = K_2 L_W + 2K_3 L_{\hat{H}}, \qquad 0 = \mu_1^{(28)} \le \mu_2^{(28)} \le \cdots \le \mu_N^{(28)}$$

Calculus image.

$$-\frac{1}{2p^2}\nabla\cdot(p^2 G_5^{-1}\nabla\phi_k^{(28)}) = \mu_k^{(28)}\phi_k^{(28)}, \qquad \mu_2^{(28)} = \min_{\substack{\phi\neq 0 \\ \int \phi\, p^2 dc = 0}} \frac{\int_{\mathcal{C}} \|\nabla\phi\|^2_{G_5^{-1}}\, p^2\, dc}{\int_{\mathcal{C}} \phi^2\, p^2\, dc}$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T2**, so the calculus image is of class **D2** (Sturm–Liouville problem and Rayleigh quotient). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the Sturm–Liouville problem and Rayleigh quotient of Lemma D2, instantiated at the point $c_{28} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D2 needs **(R1)**, **(R2)** and the standing normalisation **(N)**. Discreteness of the spectrum uses compactness of $\mathcal{C}$ and the Rellich–Kondrachov embedding $H^1 \hookrightarrow L^2$; without compactness the spectrum could be continuous and μ_2 would not be attained. The min–max formula needs self-adjointness on $L^2(p^2 dc)$, which is Θ-C(2).

(Q4) Derivation. Θ-G supplies the operator by the Gaussian-moment expansion; Θ-C(1) converts it into the bilinear form by the divergence theorem, the boundary term vanishing under the Neumann condition the kernel limit imposes. Imposing the normalisation $\int \phi^2 p^2 = 1$ and taking the first variation of $\mathcal{E}$ produces the eigenvalue equation with μ as the Lagrange multiplier; Courant–Fischer then gives the displayed variational characterisation of μ_2 on the complement of the constants.

(Q5) Retraction through the tower. Discretising the Rayleigh quotient on $\mathcal{C}_M$ returns $x^\top \mathcal{L}_{8S} x / x^\top x$, whose minimiser over mean-zero x is the 8S μ_2; the gradient becomes a difference quotient with $O(\ell^2)$ error. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \;\, 0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta_{\text{ind}}^{(28)})]$ vanishes with $\nu \ge 0$ and complementary slackness holds. By Θ-A(3) the gradient of the

novelty is $2w_{28}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: Differences of eigenvalues.

Structure: Spectral lines are eigenvalue differences, so the observable spectrum is the difference set of the Hamiltonian's spectrum. The Ritz combination principle — that line frequencies add — is exactly the statement that they come from differences of a single set.

Reduction: Lines are eigenvalue differences.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T2.

Calculus status: Exact linear; carried to the continuum by Lemma D2, and the input was algebraic, so the differential content is entirely supplied by the transform.

Also in: Astronomy.

29. Fine Structure Constant

Origin (LA edition).

$$\alpha = \frac{e^2}{4\pi\epsilon_0 \hbar c} \approx \frac{1}{137}$$

Input (8S edition).

$$y_{29}(\tau, u) = \Pi(\tau)\Big[(\Gamma W \Gamma) \otimes \mathsf{B}(\mathsf{A}_{29})\Big] z(u), \qquad z_{29}(u) = \begin{bmatrix} c_{29} \\ r_{29} u \end{bmatrix}$$

Calculus image.

$$(\mathcal{K}_\sigma \psi_{29})(c) = \int_{\mathcal{C}} \gamma(c)\, W_\sigma(c, c')\, \gamma(c')\, \mathsf{A}_{29}\, \psi_{29}(c')\, p(c')\, dc', \qquad \partial_\tau \psi_{29} = -\mathcal{L}_\star \psi_{29}, \qquad \mathcal{L}_\star = -\frac{1}{2p^2} \nabla \cdot (p^2 G_5^{-1} \nabla \cdot\,)$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T1**, so the calculus image is of class **D1** (Fredholm integral operator and its generator). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the Fredholm integral operator and its generator of Lemma D1, instantiated at the point $c_{29} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D1 needs the kernel continuous and square-integrable on $\mathcal{C} \times \mathcal{C}$, which the Gaussian form supplies unconditionally, so $\mathcal{K}_\sigma$ is Hilbert–Schmidt and hence compact. The generator statement needs **(R1)** and **(R2)**: C^4 regularity for the $O(\sigma^2)$ rate of Θ-G, and interior evaluation, since within $O(\sigma)$ of $\partial\mathcal{C}$ the expansion picks up the Neumann boundary term of Remark Θ-G″.

(Q4) Derivation. The 8S image is a Kronecker product $(\Gamma W \Gamma) \otimes \mathsf{B}(\mathsf{A})$ acting by a sum over nodes. Definition 2.2 replaces $\sum_j (\cdot)_j$ by $\int_{\mathcal{C}} (\cdot)\, p\, dc'$, turning the matrix into the displayed Fredholm operator of the second kind. Its bandwidth limit is computed by Θ-G: Taylor-expand ψ and p to second order about c, integrate against the Gaussian, use the vanishing of the odd moments and $\int w\, uu^\top du = \sigma^2 \kappa\, G_5^{-1}$, and divide by the degree. What survives at order σ^2 is $-\mathcal{L}_\star$.

(Q5) Retraction through the tower. Sampling the integral operator on $\mathcal{C}_M$ by the Riemann rule of Θ-J(1) returns the finite sum $(\Gamma W \Gamma) \otimes \mathsf{B}(\mathsf{A}_{29})$ with $O(\ell^2)$ quadrature error. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \; 0 \,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta_{\text{ind}}^{(29)})]$ vanishes with $\nu \geq 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{29}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: The dimensionless coupling scaling the perturbation.

Structure: Its smallness is why perturbation theory converges quickly in QED — the expansion parameter of the Neumann-like series in the interaction. Relativistic corrections enter at order α^2 relative to the Bohr energies.

Reduction: The constant is a perturbation expansion parameter.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T1.

Calculus status: Exact linear; carried to the continuum by Lemma D1, and the input was algebraic, so the differential content is entirely supplied by the transform.

30. Fine Structure Splitting

Origin (LA edition).

$$\Delta E \sim \alpha^2 E_n$$

Input (8S edition).

$$\mathcal{L}_{8S}\,\phi_k^{(30)} = \mu_k^{(30)}\phi_k^{(30)}, \qquad \mathcal{L}_{8S} = K_2 L_W + 2K_3 L_{\hat{H}}, \qquad 0 = \mu_1^{(30)} \leq \mu_2^{(30)} \leq \cdots \leq \mu_N^{(30)}$$

Calculus image.

$$-\frac{1}{2p^2}\nabla\cdot(p^2 G_5^{-1}\nabla\phi_k^{(30)}) = \mu_k^{(30)}\phi_k^{(30)}, \qquad \mu_2^{(30)} = \min_{\substack{\phi\neq 0 \\ \int \phi\, p^2 dc = 0}} \frac{\int_{\mathcal{C}} \|\nabla\phi\|^2_{G_5^{-1}}\, p^2\, dc}{\int_{\mathcal{C}} \phi^2\, p^2\, dc}$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T2**, so the calculus image is of class **D2** (Sturm–Liouville problem and Rayleigh quotient). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the Sturm–Liouville problem and Rayleigh quotient of Lemma D2, instantiated at the point $c_{30} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D2 needs **(R1)**, **(R2)** and the standing normalisation **(N)**. Discreteness of the spectrum uses compactness of $\mathcal{C}$ and the Rellich–Kondrachov embedding $H^1 \hookrightarrow L^2$; without compactness the spectrum could be continuous and μ_2 would not be attained. The min–max formula needs self-adjointness on $L^2(p^2 dc)$, which is Θ-C(2).

(Q4) Derivation. Θ-G supplies the operator by the Gaussian-moment expansion; Θ-C(1) converts it into the bilinear form by the divergence theorem, the boundary term vanishing under the Neumann condition the kernel limit imposes. Imposing the normalisation $\int \phi^2 p^2 = 1$ and taking the first variation of $\mathcal{E}$ produces the eigenvalue equation with μ as the Lagrange multiplier; Courant–Fischer then gives the displayed variational characterisation of μ_2 on the complement of the constants.

(Q5) Retraction through the tower. Discretising the Rayleigh quotient on $\mathcal{C}_M$ returns $x^\top \mathcal{L}_{8S} x / x^\top x$, whose minimiser over mean-zero x is the 8S μ_2; the gradient becomes a difference quotient with $O(\ell^2)$ error. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \;\, 0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta_{\text{ind}}^{(30)})]$ vanishes with $\nu \geq 0$ and complementary slackness holds. By Θ-A(3) the gradient of the

novelty is $2w_{30}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: First-order perturbation lifting a degeneracy.

Structure: The perturbation is diagonalised within each degenerate eigenspace — degenerate perturbation theory, which is a small eigenproblem inside a larger one. Splitting is the eigenvalue spread of the perturbation restricted to that subspace.

Reduction: Splitting is an eigenvalue of a restricted perturbation.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T2.

Calculus status: Exact linear; carried to the continuum by Lemma D2, and the input was algebraic, so the differential content is entirely supplied by the transform.

Angular Momentum and Spin

31. Angular Momentum Commutation

Origin (LA edition).

$$[J_i, J_j] = i\hbar\epsilon_{ijk}J_k$$

Input (8S edition).

$$y_{31}(\tau, u) = \Pi(\tau)\Big[(\Gamma W\Gamma) \otimes \mathsf{B}(\mathsf{A}_{31})\Big] z(u), \qquad z_{31}(u) = \begin{bmatrix} c_{31} \\ r_{31}u \end{bmatrix}$$

Calculus image.

$$(\mathcal{K}_\sigma\psi_{31})(c) = \int_{\mathcal{C}} \gamma(c)\, W_\sigma(c, c')\, \gamma(c')\, \mathsf{A}_{31}\, \psi_{31}(c')\, p(c')\, dc', \qquad \partial_\tau\psi_{31} = -\mathcal{L}_\star\psi_{31}, \qquad \mathcal{L}_\star = -\frac{1}{2p^2}\nabla\cdot(p^2 G_5^{-1}\nabla\,\cdot\,)$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T1**, so the calculus image is of class **D1** (Fredholm integral operator and its generator). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the Fredholm integral operator and its generator of Lemma D1, instantiated at the point $c_{31} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D1 needs the kernel continuous and square-integrable on $\mathcal{C} \times \mathcal{C}$, which the Gaussian form supplies unconditionally, so $\mathcal{K}_\sigma$ is Hilbert–Schmidt and hence compact. The generator statement needs **(R1)** and **(R2)**: C^4 regularity for the $O(\sigma^2)$ rate of Θ-G, and interior evaluation, since within $O(\sigma)$ of $\partial\mathcal{C}$ the expansion picks up the Neumann boundary term of Remark Θ-G″.

(Q4) Derivation. The 8S image is a Kronecker product $(\Gamma W\Gamma) \otimes \mathsf{B}(\mathsf{A})$ acting by a sum over nodes. Definition 2.2 replaces $\sum_j(\cdot)_j$ by $\int_{\mathcal{C}}(\cdot)\, p\, dc'$, turning the matrix into the displayed Fredholm operator of the second kind. Its bandwidth limit is computed by Θ-G: Taylor-expand ψ and p to second order about c, integrate against the Gaussian, use the vanishing of the odd moments and $\int w\, uu^\top du = \sigma^2\kappa\, G_5^{-1}$, and divide by the degree. What survives at order σ^2 is $-\mathcal{L}_\star$.

(Q5) Retraction through the tower. Sampling the integral operator on $\mathcal{C}_M$ by the Riemann rule of Θ-J(1) returns the finite sum $(\Gamma W\Gamma) \otimes \mathsf{B}(\mathsf{A}_{31})$ with $O(\ell^2)$ quadrature error. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \; 0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta_{\text{ind}}^{(31)})]$ vanishes with $\nu \geq 0$ and complementary slackness holds. By Θ-A(3) the gradient of the

novelty is $2w_{31}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: The $\mathfrak{su}(2)$ Lie algebra structure constants.

Structure: The entire theory of angular momentum follows from these three commutators — the representation theory of $\mathfrak{su}(2)$. The Casimir J^2 commutes with all of them, so it labels irreducible representations.

Reduction: None needed.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T1.

Calculus status: Exact linear; carried to the continuum by Lemma D1, and the input was algebraic, so the differential content is entirely supplied by the transform.

Also in: Abstract Algebra.

32. Angular Momentum Eigenvalues

Origin (LA edition).

$$J^2\,|jm\rangle = \hbar^2 j(j+1)\,|jm\rangle\,, \qquad J_z\,|jm\rangle = \hbar m\,|jm\rangle$$

Input (8S edition).

$$\mathcal{L}_{8S}\,\phi_k^{(32)} = \mu_k^{(32)}\phi_k^{(32)}, \qquad \mathcal{L}_{8S} = K_2 L_W + 2K_3 L_{\hat{H}}, \qquad 0 = \mu_1^{(32)} \le \mu_2^{(32)} \le \cdots \le \mu_N^{(32)}$$

Calculus image.

$$-\frac{1}{2p^2}\nabla\cdot(p^2 G_5^{-1}\nabla\phi_k^{(32)}) = \mu_k^{(32)}\phi_k^{(32)}, \qquad \mu_2^{(32)} = \min_{\substack{\phi\neq 0 \\ \int \phi\, p^2\, dc = 0}} \frac{\int_{\mathcal{C}} \|\nabla\phi\|^2_{G_5^{-1}}\, p^2\, dc}{\int_{\mathcal{C}} \phi^2\, p^2\, dc}$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T2**, so the calculus image is of class **D2** (Sturm–Liouville problem and Rayleigh quotient). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the Sturm–Liouville problem and Rayleigh quotient of Lemma D2, instantiated at the point $c_{32} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D2 needs **(R1)**, **(R2)** and the standing normalisation **(N)**. Discreteness of the spectrum uses compactness of $\mathcal{C}$ and the Rellich–Kondrachov embedding $H^1 \hookrightarrow L^2$; without compactness the spectrum could be continuous and μ_2 would not be attained. The min–max formula needs self-adjointness on $L^2(p^2 dc)$, which is Θ-C(2).

(Q4) Derivation. Θ-G supplies the operator by the Gaussian-moment expansion; Θ-C(1) converts it into the bilinear form by the divergence theorem, the boundary term vanishing under the Neumann condition the kernel limit imposes. Imposing the normalisation $\int \phi^2 p^2 = 1$ and taking the first variation of $\mathcal{E}$ produces the eigenvalue equation with μ as the Lagrange multiplier; Courant–Fischer then gives the displayed variational characterisation of μ_2 on the complement of the constants.

(Q5) Retraction through the tower. Discretising the Rayleigh quotient on $\mathcal{C}_M$ returns $x^\top \mathcal{L}_{8S} x / x^\top x$, whose minimiser over mean-zero x is the 8S μ_2; the gradient becomes a difference quotient with $O(\ell^2)$ error. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\,I_3\ 0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta_{\text{ind}}^{(32)})]$ vanishes with $\nu \ge 0$ and complementary slackness holds. By Θ-A(3) the gradient of the

novelty is $2w_{32}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: Simultaneous eigenvectors of two commuting operators.

Structure: $[J^2, J_z] = 0$ permits a common eigenbasis, and the ladder operators $J_\pm$ generate the whole $(2j+1)$-dimensional multiplet from any one member. The dimension is the representation's dimension — degeneracy is representation theory.

Reduction: None needed.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T2.

Calculus status: Exact linear; carried to the continuum by Lemma D2, and the input was algebraic, so the differential content is entirely supplied by the transform.

33. Spin-1/2 Pauli Matrices

Origin (LA edition).

$$\sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

Input (8S edition).

$$y_{33}(\tau, u) = \Pi(\tau)\Big[(\Gamma W \Gamma) \otimes \mathsf{B}(\mathsf{A}_{33})\Big] z(u), \qquad z_{33}(u) = \begin{bmatrix} c_{33} \\ r_{33} u \end{bmatrix}$$

Calculus image.

$$(\mathcal{K}_\sigma \psi_{33})(c) = \int_{\mathcal{C}} \gamma(c)\, W_\sigma(c, c')\, \gamma(c')\, \mathsf{A}_{33}\, \psi_{33}(c')\, p(c')\, dc', \qquad \partial_\tau \psi_{33} = -\mathcal{L}_\star \psi_{33}, \qquad \mathcal{L}_\star = -\frac{1}{2p^2} \nabla \cdot (p^2 G_5^{-1} \nabla \cdot\,)$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T1**, so the calculus image is of class **D1** (Fredholm integral operator and its generator). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the Fredholm integral operator and its generator of Lemma D1, instantiated at the point $c_{33} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D1 needs the kernel continuous and square-integrable on $\mathcal{C} \times \mathcal{C}$, which the Gaussian form supplies unconditionally, so $\mathcal{K}_\sigma$ is Hilbert–Schmidt and hence compact. The generator statement needs **(R1)** and **(R2)**: C^4 regularity for the $O(\sigma^2)$ rate of Θ-G, and interior evaluation, since within $O(\sigma)$ of $\partial\mathcal{C}$ the expansion picks up the Neumann boundary term of Remark Θ-G″.

(Q4) Derivation. The 8S image is a Kronecker product $(\Gamma W \Gamma) \otimes \mathsf{B}(\mathsf{A})$ acting by a sum over nodes. Definition 2.2 replaces $\sum_j (\cdot)_j$ by $\int_{\mathcal{C}} (\cdot)\, p\, dc'$, turning the matrix into the displayed Fredholm operator of the second kind. Its bandwidth limit is computed by Θ-G: Taylor-expand ψ and p to second order about c, integrate against the Gaussian, use the vanishing of the odd moments and $\int w\, uu^\top du = \sigma^2 \kappa\, G_5^{-1}$, and divide by the degree. What survives at order σ^2 is $-\mathcal{L}_\star$.

(Q5) Retraction through the tower. Sampling the integral operator on $\mathcal{C}_M$ by the Riemann rule of Θ-J(1) returns the finite sum $(\Gamma W \Gamma) \otimes \mathsf{B}(\mathsf{A}_{33})$ with $O(\ell^2)$ quadrature error. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \; 0 \,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta_{\text{ind}}^{(33)})]$ vanishes with $\nu \geq 0$ and complementary slackness holds. By Θ-A(3) the gradient of the

novelty is $2w_{33}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: Explicit 2×2 Hermitian traceless matrices.

Structure: Together with I they span the Hermitian 2×2 matrices, so every qubit observable is a real combination. $\sigma_i^2 = I$, $\sigma_i\sigma_j = i\epsilon_{ijk}\sigma_k$, and $e^{-i\theta\hat{\mathbf{n}}\cdot\sigma/2}$ is a rotation with 4π periodicity — the $SU(2)$ double cover.

Reduction: None needed.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T1.

Calculus status: Exact linear; carried to the continuum by Lemma D1, and the input was algebraic, so the differential content is entirely supplied by the transform.

34. Spin Magnetic Moment

Origin (LA edition).

$$\mu = -g\frac{\mu_B}{\hbar}\mathbf{S}$$

Input (8S edition).

$$\mathcal{L}_{8S}\,\phi_k^{(34)} = \mu_k^{(34)}\phi_k^{(34)}, \qquad \mathcal{L}_{8S} = K_2 L_W + 2K_3 L_{\hat{H}}, \qquad 0 = \mu_1^{(34)} \le \mu_2^{(34)} \le \cdots \le \mu_N^{(34)}$$

Calculus image.

$$-\frac{1}{2p^2}\nabla\cdot(p^2 G_5^{-1}\nabla\phi_k^{(34)}) = \mu_k^{(34)}\phi_k^{(34)}, \qquad \mu_2^{(34)} = \min_{\substack{\phi\neq0\\ \int\phi\,p^2dc=0}} \frac{\int_{\mathcal{C}} \|\nabla\phi\|^2_{G_5^{-1}}\,p^2\,dc}{\int_{\mathcal{C}}\phi^2\,p^2\,dc}$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T2**, so the calculus image is of class **D2** (Sturm–Liouville problem and Rayleigh quotient). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the Sturm–Liouville problem and Rayleigh quotient of Lemma D2, instantiated at the point $c_{34} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D2 needs **(R1)**, **(R2)** and the standing normalisation **(N)**. Discreteness of the spectrum uses compactness of $\mathcal{C}$ and the Rellich–Kondrachov embedding $H^1 \hookrightarrow L^2$; without compactness the spectrum could be continuous and μ_2 would not be attained. The min–max formula needs self-adjointness on $L^2(p^2dc)$, which is Θ-C(2).

(Q4) Derivation. Θ-G supplies the operator by the Gaussian-moment expansion; Θ-C(1) converts it into the bilinear form by the divergence theorem, the boundary term vanishing under the Neumann condition the kernel limit imposes. Imposing the normalisation $\int\phi^2p^2 = 1$ and taking the first variation of $\mathcal{E}$ produces the eigenvalue equation with μ as the Lagrange multiplier; Courant–Fischer then gives the displayed variational characterisation of μ_2 on the complement of the constants.

(Q5) Retraction through the tower. Discretising the Rayleigh quotient on $\mathcal{C}_M$ returns $x^\top\mathcal{L}_{8S}x/x^\top x$, whose minimiser over mean-zero x is the 8S μ_2; the gradient becomes a difference quotient with $O(\ell^2)$ error. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\,I_3\ 0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta_{\text{ind}}^{(34)})]$ vanishes with $\nu \ge 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{34}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: An operator proportional to the spin vector operator.

Structure: Proportionality means the moment shares eigenvectors with the spin, so measuring one measures the other. $g \approx 2$ from the Dirac equation, with QED corrections making it the most precisely tested quantity in physics.

Reduction: None needed.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T2.

Calculus status: Exact linear; carried to the continuum by Lemma D2, and the input was algebraic, so the differential content is entirely supplied by the transform.

35. Zeeman Effect

Origin (LA edition).

$$H' = -\mu \cdot \mathbf{B} = \frac{g\mu_B}{\hbar} B\, S_z$$

Input (8S edition).

$$\mathcal{L}_{8S}\, \phi_k^{(35)} = \mu_k^{(35)} \phi_k^{(35)}, \qquad \mathcal{L}_{8S} = K_2 L_W + 2K_3 L_{\hat{H}}, \qquad 0 = \mu_1^{(35)} \le \mu_2^{(35)} \le \cdots \le \mu_N^{(35)}$$

Calculus image.

$$-\frac{1}{2p^2} \nabla \cdot (p^2 G_5^{-1} \nabla \phi_k^{(35)}) = \mu_k^{(35)} \phi_k^{(35)}, \qquad \mu_2^{(35)} = \min_{\substack{\phi \neq 0 \\ \int \phi\, p^2 dc = 0}} \frac{\int_{\mathcal{C}} \|\nabla \phi\|^2_{G_5^{-1}}\, p^2\, dc}{\int_{\mathcal{C}} \phi^2\, p^2\, dc}$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T2**, so the calculus image is of class **D2** (Sturm–Liouville problem and Rayleigh quotient). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the Sturm–Liouville problem and Rayleigh quotient of Lemma D2, instantiated at the point $c_{35} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D2 needs **(R1)**, **(R2)** and the standing normalisation **(N)**. Discreteness of the spectrum uses compactness of $\mathcal{C}$ and the Rellich–Kondrachov embedding $H^1 \hookrightarrow L^2$; without compactness the spectrum could be continuous and μ_2 would not be attained. The min–max formula needs self-adjointness on $L^2(p^2 dc)$, which is Θ-C(2).

(Q4) Derivation. Θ-G supplies the operator by the Gaussian-moment expansion; Θ-C(1) converts it into the bilinear form by the divergence theorem, the boundary term vanishing under the Neumann condition the kernel limit imposes. Imposing the normalisation $\int \phi^2 p^2 = 1$ and taking the first variation of $\mathcal{E}$ produces the eigenvalue equation with μ as the Lagrange multiplier; Courant–Fischer then gives the displayed variational characterisation of μ_2 on the complement of the constants.

(Q5) Retraction through the tower. Discretising the Rayleigh quotient on $\mathcal{C}_M$ returns $x^\top \mathcal{L}_{8S} x / x^\top x$, whose minimiser over mean-zero x is the 8S μ_2; the gradient becomes a difference quotient with $O(\ell^2)$ error. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \;\, 0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta_{\text{ind}}^{(35)})]$ vanishes with $\nu \ge 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{35}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: A diagonal perturbation in the J_z eigenbasis.

Structure: The field breaks rotational symmetry, so the degeneracy of the multiplet lifts — splitting is the eigenvalue spread of a diagonal perturbation. Symmetry breaking is degeneracy lifting, stated spectrally.

Reduction: None needed.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T2.

Calculus status: Exact linear; carried to the continuum by Lemma D2, and the input was algebraic, so the differential content is entirely supplied by the transform.

36. Landé g-Factor

Origin (LA edition).

$$g_J = 1 + \frac{j(j+1) + s(s+1) - \ell(\ell+1)}{2j(j+1)}$$

Input (8S edition).

$$P_5^{(36)} = C_5 C_5^{+} = P_4 + \frac{w_{36} w_{36}^{\top}}{\|w_{36}\|_2^2}, \qquad w_{36} = (I - C_4 C_4^{+}) e_{36}, \qquad \eta_{\text{ind}}^{(36)} = \frac{\|w_{36}\|_2^2}{\|e_{36}\|_2^2 + \varepsilon}$$

Calculus image.

$$\langle e_{36} - P_{\mathcal{V}} e_{36},\ v \rangle = 0 \quad \forall v \in \mathcal{V}, \qquad \eta_{\text{ind}}^{(36)} = \frac{\det \mathrm{Gr}(C_4, e_{36})}{\det \mathrm{Gr}(C_4)(\|e_{36}\|^2 + \varepsilon)}, \qquad \nabla_{e_{36}} \|w_{36}\|^2 = 2\, w_{36}$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T4**, so the calculus image is of class **D4** (Hilbert projection theorem and Gram volume). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the Hilbert projection theorem and Gram volume of Lemma D4, instantiated at the point $c_{36} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D4 needs C_4 of full column rank, so that $\mathrm{Gr}(C_4)$ is invertible and the Schur determinant formula applies. Completeness of $\mathcal{V}$ is needed for the Hilbert projection theorem; a finite-dimensional span is automatically closed, so this costs nothing here.

(Q4) Derivation. Differentiate the least-squares functional: $\frac{d}{ds}\|e - Pe - sv\|^2\big|_{s=0} = -2\langle e - Pe, v\rangle$, and vanishing for all v is the orthogonality condition. For the determinant form, apply the Schur formula to the block Gram matrix, whose Schur complement is $\langle e, (I - P_{\mathcal{V}})e\rangle = \|w\|^2$ by idempotence and self-adjointness of $I - P_{\mathcal{V}}$. The gradient follows by symmetry of $I - P$.

(Q5) Retraction through the tower. On the lattice the inner products become finite sums, the Gram determinants become the matrix Gram determinants, and the ratio becomes the 8S residual $\|w_{36}\|^2/(\|e_{36}\|^2 + \varepsilon)$. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \ 0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta_{\text{ind}}^{(36)})]$ vanishes with $\nu \geq 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{36}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: A projection coefficient from the Wigner–Eckart theorem.

Structure: The magnetic moment is not parallel to **J**, so only its projection onto **J** survives — the g-factor is exactly that projection coefficient. It is a matrix element ratio.

Reduction: The factor is a projection coefficient.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T4.

Calculus status: Exact linear; carried to the continuum by Lemma D4, and the input was algebraic, so the differential content is entirely supplied by the transform.

37. Spin–Orbit Coupling

Origin (LA edition).

$$H_{SO} = \xi(r)\,\mathbf{L}\cdot\mathbf{S}$$

Input (8S edition).

$$y_{37}(\tau,u) = \Pi(\tau)\Big[(\Gamma W\Gamma)\otimes \mathsf{B}(\mathsf{A}_{37})\Big]z(u), \qquad z_{37}(u) = \begin{bmatrix} c_{37} \\ r_{37}u \end{bmatrix}$$

Calculus image.

$$(\mathcal{K}_\sigma\psi_{37})(c) = \int_{\mathcal{C}} \gamma(c)\,W_\sigma(c,c')\,\gamma(c')\,\mathsf{A}_{37}\,\psi_{37}(c')\,p(c')\,dc', \qquad \partial_\tau\psi_{37} = -\mathcal{L}_\star\psi_{37}, \qquad \mathcal{L}_\star = -\frac{1}{2p^2}\nabla\cdot(p^2G_5^{-1}\nabla\,\cdot\,)$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T1**, so the calculus image is of class **D1** (Fredholm integral operator and its generator). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the Fredholm integral operator and its generator of Lemma D1, instantiated at the point $c_{37} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D1 needs the kernel continuous and square-integrable on $\mathcal{C}\times\mathcal{C}$, which the Gaussian form supplies unconditionally, so $\mathcal{K}_\sigma$ is Hilbert–Schmidt and hence compact. The generator statement needs **(R1)** and **(R2)**: C^4 regularity for the $O(\sigma^2)$ rate of Θ-G, and interior evaluation, since within $O(\sigma)$ of $\partial\mathcal{C}$ the expansion picks up the Neumann boundary term of Remark Θ-G″.

(Q4) Derivation. The 8S image is a Kronecker product $(\Gamma W\Gamma)\otimes\mathsf{B}(\mathsf{A})$ acting by a sum over nodes. Definition 2.2 replaces $\sum_j(\cdot)_j$ by $\int_{\mathcal{C}}(\cdot)\,p\,dc'$, turning the matrix into the displayed Fredholm operator of the second kind. Its bandwidth limit is computed by Θ-G: Taylor-expand ψ and p to second order about c, integrate against the Gaussian, use the vanishing of the odd moments and $\int w\,uu^\top du = \sigma^2\kappa\,G_5^{-1}$, and divide by the degree. What survives at order σ^2 is $-\mathcal{L}_\star$.

(Q5) Retraction through the tower. Sampling the integral operator on $\mathcal{C}_M$ by the Riemann rule of Θ-J(1) returns the finite sum $(\Gamma W\Gamma)\otimes\mathsf{B}(\mathsf{A}_{37})$ with $O(\ell^2)$ quadrature error. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma\to I$, $W\to I$, $\mathcal{R}\to 0$, $K_2,K_3,J_2,J_3\to 0$ and $\Pi\to[\,I_3\;0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\mathrm{ind}} - \eta_{\mathrm{ind}}^{(37)})]$ vanishes with $\nu\ge 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{37}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: A bilinear coupling of two vector operators.

Structure: $\mathbf{L}\cdot\mathbf{S} = \frac{1}{2}(J^2 - L^2 - S^2)$, so it is diagonal in the coupled $|j\ell s\rangle$ basis — the coupling *selects* which basis diagonalises the Hamiltonian. Changing from uncoupled to coupled basis is a Clebsch–Gordan change of basis.

Reduction: None needed.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T1.

Calculus status: Exact linear; carried to the continuum by Lemma D1, and the input was algebraic, so the differential content is entirely supplied by the transform.

38. Clebsch–Gordan Addition

Origin (LA edition).

$$|j_1 j_2; JM\rangle = \sum_{m_1 m_2} C^{JM}_{j_1 m_1 j_2 m_2} \, |j_1 m_1\rangle \, |j_2 m_2\rangle$$

Input (8S edition).

$$y_{38}(\tau, u) = \Pi(\tau)\Big[(\Gamma W \Gamma) \otimes \mathsf{B}(\mathsf{A}_{38})\Big] z(u), \qquad z_{38}(u) = \begin{bmatrix} c_{38} \\ r_{38} u \end{bmatrix}$$

Calculus image.

$$(\mathcal{K}_\sigma \psi_{38})(c) = \int_{\mathcal{C}} \gamma(c)\, W_\sigma(c, c')\, \gamma(c')\, \mathsf{A}_{38}\, \psi_{38}(c')\, p(c')\, dc', \qquad \partial_\tau \psi_{38} = -\mathcal{L}_\star \psi_{38}, \qquad \mathcal{L}_\star = -\frac{1}{2p^2} \nabla \cdot (p^2 G_5^{-1} \nabla \cdot\,)$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T1**, so the calculus image is of class **D1** (Fredholm integral operator and its generator). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the Fredholm integral operator and its generator of Lemma D1, instantiated at the point $c_{38} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D1 needs the kernel continuous and square-integrable on $\mathcal{C} \times \mathcal{C}$, which the Gaussian form supplies unconditionally, so $\mathcal{K}_\sigma$ is Hilbert–Schmidt and hence compact. The generator statement needs **(R1)** and **(R2)**: C^4 regularity for the $O(\sigma^2)$ rate of Θ-G, and interior evaluation, since within $O(\sigma)$ of $\partial\mathcal{C}$ the expansion picks up the Neumann boundary term of Remark Θ-G″.

(Q4) Derivation. The 8S image is a Kronecker product $(\Gamma W \Gamma) \otimes \mathsf{B}(\mathsf{A})$ acting by a sum over nodes. Definition 2.2 replaces $\sum_j (\cdot)_j$ by $\int_{\mathcal{C}} (\cdot)\, p\, dc'$, turning the matrix into the displayed Fredholm operator of the second kind. Its bandwidth limit is computed by Θ-G: Taylor-expand ψ and p to second order about c, integrate against the Gaussian, use the vanishing of the odd moments and $\int w\, uu^\top du = \sigma^2 \kappa\, G_5^{-1}$, and divide by the degree. What survives at order σ^2 is $-\mathcal{L}_\star$.

(Q5) Retraction through the tower. Sampling the integral operator on $\mathcal{C}_M$ by the Riemann rule of Θ-J(1) returns the finite sum $(\Gamma W \Gamma) \otimes \mathsf{B}(\mathsf{A}_{38})$ with $O(\ell^2)$ quadrature error. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \; 0 \,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta^{(38)}_{\text{ind}})]$ vanishes with $\nu \geq 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{38}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: A unitary change of basis between two decompositions of a tensor product.

Structure: $\mathbf{j}_1 \otimes \mathbf{j}_2 = \bigoplus_J \mathbf{J}$ — the tensor product of two irreducible representations decomposes into a direct sum. The Clebsch–Gordan coefficients are the entries of the unitary matrix effecting that block-diagonalisation.

Reduction: None needed.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T1.

Calculus status: Exact linear; carried to the continuum by Lemma D1, and the input was algebraic, so the differential content is entirely supplied by the transform.

39. Wigner–Eckart Theorem

Origin (LA edition).

$$\langle j'm' |\, T^k_q \,| jm \rangle = \langle jm; kq \mid j'm' \rangle \, \frac{\langle j' \| T^k \| j \rangle}{\sqrt{2j'+1}}$$

Input (8S edition).

$$\operatorname{rank}(C_5) = \operatorname{rank}(C_4) + \mathbf{1}[\eta^{(39)}_{\text{ind}} > 0], \qquad \dim\ker \mathcal{L}_{8S} = \#\operatorname{comp}(W + \hat{H})$$

Calculus image.

$$\dim\ker \mathcal{L}_\star = \dim H^0(\mathcal{C}) = \#\{\text{components}\}, \qquad \operatorname{ind} \mathcal{L}_\star = 0, \qquad \det \operatorname{Gr}(C_4, e_{39}) \neq 0 \iff \text{rank increases}$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T5**, so the calculus image is of class **D5** (kernel dimension, index, and H^0). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the kernel dimension, index, and H^0 of Lemma D5, instantiated at the point $c_{39} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D5 needs $\mathcal{L}_\star$ self-adjoint with compact resolvent, which is Θ-G under **(R1)**–**(R2)**; the identification of ker with locally constant functions needs $G_5 \succ 0$ and $p > 0$, both standing. The index vanishes because the operator is self-adjoint, so coker ≅ ker.

(Q4) Derivation. Set $\mathcal{E}[\phi] = 0$ in Θ-C(1). Since $G_5^{-1} \succ 0$ and $p > 0$, this forces $\nabla\phi = 0$ almost everywhere, hence ϕ locally constant, hence $\ker \mathcal{L}_\star$ spanned by the indicator functions of the connected components — which is precisely H^0. Self-adjointness gives coker ≅ ker and therefore index zero.

(Q5) Retraction through the tower. On the lattice $\dim\ker \mathcal{L}_\star$ becomes $\dim\ker \mathcal{L}_{8S}$, the component count of the coupling graph, and the Gram condition becomes the 8S rank increment. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \; 0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta^{(39)}_{\text{ind}})]$ vanishes with $\nu \geq 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{39}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: Factorisation of a matrix element into geometry and dynamics.

Structure: All m-dependence is in the Clebsch–Gordan coefficient; the physics is in one reduced matrix element. This is a rank-one factorisation of the matrix of matrix elements — a huge compression forced entirely by symmetry.

Reduction: The theorem is a rank-one factorisation.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T5.

Calculus status: Exact linear; carried to the continuum by Lemma D5, and the input was algebraic, so the differential content is entirely supplied by the transform.

40. Electric Dipole Selection Rules

Origin (LA edition).

$$\Delta\ell = \pm 1, \qquad \Delta m = 0, \pm 1$$

Input (8S edition).

$$\widehat{P}^{(40)} = D^{-1}_{\Gamma W \Gamma} \Gamma W \Gamma, \qquad \pi^{8S} \propto \Gamma W \Gamma \mathbf{1}, \qquad \pi^{8S}_{40} \widehat{P}^{(40)}_{40j} = \pi^{8S}_{j} \widehat{P}^{(40)}_{j40}$$

Calculus image.

$$\partial_t \rho_{40} = \nabla \cdot \Big(G_5^{-1} (\tfrac{1}{2} \nabla \rho_{40} + \rho_{40} \nabla V) \Big), \qquad \rho_\infty \propto e^{-2V}, \qquad \partial_t \rho_{40} = \nabla \cdot \Big(\rho_{40} \, \nabla \frac{\delta \mathcal{F}}{\delta \rho} \Big)$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T10**, so the calculus image is of class **D10** (Fokker–Planck equation and Wasserstein gradient flow). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the Fokker–Planck equation and Wasserstein gradient flow of Lemma D10, instantiated at the point $c_{40} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D10 needs the chain to be reversible, which holds because $\Gamma W \Gamma$ is symmetric, and needs $V = -\frac{1}{2} \log \pi^{8S}$ to be C^1, i.e. the stationary density bounded away from zero. The Wasserstein gradient-flow form needs $\mathcal{F}$ displacement-convex, which holds when V is convex.

(Q4) Derivation. The 8S chain has generator $\widehat{P} - I$; its continuum limit is the adjoint of $\mathcal{L}_\star$ by Θ-G. Writing the reversible measure as $\pi^{8S} \propto e^{-2V}$ and expanding the divergence puts the generator in drift–diffusion form. Stationarity is verified by substitution: $\frac{1}{2} \nabla e^{-2V} + e^{-2V} \nabla V = -e^{-2V} \nabla V + e^{-2V} \nabla V = 0$. The gradient-flow form is the Jordan–Kinderlehrer–Otto identification of Fokker–Planck as steepest descent of $\mathcal{F}[\rho] = \int (\frac{1}{2} \rho \log \rho + \rho V)$ in the Wasserstein-2 metric.

(Q5) Retraction through the tower. Discretising the Fokker–Planck operator on $\mathcal{C}_M$ returns the generator $\widehat{P} - I$ of the 8S reversible chain, with ρ_∞ becoming π^{8S}. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \;\, 0 \,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta^{(40)}_{\text{ind}})]$ vanishes with $\nu \geq 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{40}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: Vanishing matrix elements forced by symmetry.

Structure: $\langle f | \, \hat{\mathbf{r}} \, | i \rangle = 0$ unless the Clebsch–Gordan coefficient is nonzero — selection rules are a sparsity pattern in the transition matrix imposed by group theory. Most entries are exactly zero, which is why atomic spectra are sparse.

Reduction: Selection rules are a symmetry-forced sparsity pattern.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T10.

Calculus status: Exact linear; carried to the continuum by Lemma D10, and the input was algebraic, so the differential content is entirely supplied by the transform.

Approximation Methods and Scattering

41. Time-Independent Perturbation Theory

Origin (LA edition).

$$E_n^{(1)} = \langle n|\, H' \,|n\rangle\,, \qquad E_n^{(2)} = \sum_{m\neq n} \frac{\left|\langle m|\, H' \,|n\rangle\right|^2}{E_n - E_m}$$

Input (8S edition).

$$\mathcal{L}_{8S}\,\phi_k^{(41)} = \mu_k^{(41)}\phi_k^{(41)}, \qquad \mathcal{L}_{8S} = K_2 L_W + 2K_3 L_{\hat{H}}, \qquad 0 = \mu_1^{(41)} \le \mu_2^{(41)} \le \cdots \le \mu_N^{(41)}$$

Calculus image.

$$-\frac{1}{2p^2}\nabla\cdot(p^2 G_5^{-1}\nabla\phi_k^{(41)}) = \mu_k^{(41)}\phi_k^{(41)}, \qquad \mu_2^{(41)} = \min_{\substack{\phi\neq 0 \\ \int \phi\, p^2 dc = 0}} \frac{\int_{\mathcal{C}} \|\nabla\phi\|^2_{G_5^{-1}}\, p^2\, dc}{\int_{\mathcal{C}} \phi^2\, p^2\, dc}$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T2**, so the calculus image is of class **D2** (Sturm–Liouville problem and Rayleigh quotient). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the Sturm–Liouville problem and Rayleigh quotient of Lemma D2, instantiated at the point $c_{41} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D2 needs **(R1)**, **(R2)** and the standing normalisation **(N)**. Discreteness of the spectrum uses compactness of $\mathcal{C}$ and the Rellich–Kondrachov embedding $H^1 \hookrightarrow L^2$; without compactness the spectrum could be continuous and μ_2 would not be attained. The min–max formula needs self-adjointness on $L^2(p^2 dc)$, which is Θ-C(2).

(Q4) Derivation. Θ-G supplies the operator by the Gaussian-moment expansion; Θ-C(1) converts it into the bilinear form by the divergence theorem, the boundary term vanishing under the Neumann condition the kernel limit imposes. Imposing the normalisation $\int \phi^2 p^2 = 1$ and taking the first variation of $\mathcal{E}$ produces the eigenvalue equation with μ as the Lagrange multiplier; Courant–Fischer then gives the displayed variational characterisation of μ_2 on the complement of the constants.

(Q5) Retraction through the tower. Discretising the Rayleigh quotient on $\mathcal{C}_M$ returns $x^\top \mathcal{L}_{8S} x / x^\top x$, whose minimiser over mean-zero x is the 8S μ_2; the gradient becomes a difference quotient with $O(\ell^2)$ error. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \;\, 0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta_{\text{ind}}^{(41)})]$ vanishes with $\nu \ge 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{41}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: Eigenvalue perturbation of a Hermitian matrix.

Structure: Exactly the standard first- and second-order eigenvalue perturbation formulas. The energy denominators are eigenvalue gaps, so the expansion fails when levels are close — near-degeneracy is ill-conditioning of the eigenproblem.

Reduction: None needed.

Status: Exact linear (perturbation of a Hermitian eigenproblem).

8S status: Exact linear (perturbation of a Hermitian eigenproblem); preserved under the transform by Theorem C and Lemma T2.

Calculus status: Exact linear (perturbation of a Hermitian eigenproblem); carried to the continuum by Lemma D2, and the input was algebraic, so the differential content is entirely supplied by the transform.

42. Degenerate Perturbation Theory

Origin (LA edition).

$$\det(H' - E^{(1)} I)_{\text{degenerate block}} = 0$$

Input (8S edition).

$$\mathcal{L}_{8S}\,\phi_k^{(42)} = \mu_k^{(42)}\phi_k^{(42)}, \qquad \mathcal{L}_{8S} = K_2 L_W + 2K_3 L_{\hat{H}}, \qquad 0 = \mu_1^{(42)} \le \mu_2^{(42)} \le \cdots \le \mu_N^{(42)}$$

Calculus image.

$$-\frac{1}{2p^2}\nabla\cdot(p^2 G_5^{-1}\nabla\phi_k^{(42)}) = \mu_k^{(42)}\phi_k^{(42)}, \qquad \mu_2^{(42)} = \min_{\substack{\phi\neq 0 \\ \int \phi\, p^2 dc = 0}} \frac{\int_{\mathcal{C}} \|\nabla\phi\|^2_{G_5^{-1}}\, p^2\, dc}{\int_{\mathcal{C}} \phi^2\, p^2\, dc}$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T2**, so the calculus image is of class **D2** (Sturm–Liouville problem and Rayleigh quotient). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the Sturm–Liouville problem and Rayleigh quotient of Lemma D2, instantiated at the point $c_{42} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D2 needs **(R1)**, **(R2)** and the standing normalisation **(N)**. Discreteness of the spectrum uses compactness of $\mathcal{C}$ and the Rellich–Kondrachov embedding $H^1 \hookrightarrow L^2$; without compactness the spectrum could be continuous and μ_2 would not be attained. The min–max formula needs self-adjointness on $L^2(p^2 dc)$, which is Θ-C(2).

(Q4) Derivation. Θ-G supplies the operator by the Gaussian-moment expansion; Θ-C(1) converts it into the bilinear form by the divergence theorem, the boundary term vanishing under the Neumann condition the kernel limit imposes. Imposing the normalisation $\int \phi^2 p^2 = 1$ and taking the first variation of $\mathcal{E}$ produces the eigenvalue equation with μ as the Lagrange multiplier; Courant–Fischer then gives the displayed variational characterisation of μ_2 on the complement of the constants.

(Q5) Retraction through the tower. Discretising the Rayleigh quotient on $\mathcal{C}_M$ returns $x^\top \mathcal{L}_{8S} x / x^\top x$, whose minimiser over mean-zero x is the 8S μ_2; the gradient becomes a difference quotient with $O(\ell^2)$ error. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \;\, 0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta_{\text{ind}}^{(42)})]$ vanishes with $\nu \ge 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{42}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: A small eigenproblem within the degenerate eigenspace.

Structure: When the gap vanishes, the naive expansion diverges; the fix is to diagonalise H' restricted to the degenerate subspace first. This is exactly how one handles a defective or near-defective eigenproblem numerically.

Reduction: None needed.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T2.

Calculus status: Exact linear; carried to the continuum by Lemma D2, and the input was algebraic, so the differential content is entirely supplied by the transform.

43. Variational Principle

Origin (LA edition).

$$E_0 \le \frac{\langle\psi|\, H\,|\psi\rangle}{\langle\psi\mid\psi\rangle}$$

Input (8S edition).

$$E_{43}^{8S}(z) = z^\top \mathcal{G}_8 z\Big|_{\text{node }43}, \qquad \big\|z_{43}(u) - z_j(u)\big\|_{\mathcal{G}_8} = \delta_{43j}^{(8)} = \sqrt{(c_{43}-c_j)^\top G_5 (c_{43}-c_j) + (r_{43}-r_j)^2}$$

Calculus image.

$$\mathcal{E}_{43}[\phi] = \tfrac{1}{2}\int_{\mathcal{C}} \|\nabla\phi\|^2_{G_5^{-1}}\, p^2\, dc, \qquad \delta\mathcal{E}_{43} = 0 \iff \mathcal{L}_\star\phi = 0, \qquad ds_{43}^2 = dc^\top(G_5 + \nabla r\,\nabla r^\top)dc + r_{43}^2\, d\Omega_3^2$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T3**, so the calculus image is of class **D3** (Dirichlet energy and the cone metric). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the Dirichlet energy and the cone metric of Lemma D3, instantiated at the point $c_{43} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D3 needs **(R3)**, $r \in C^1$ with $r > 0$, for the bundle to be a smooth manifold; at $r = 0$ the fibre collapses and ds^2 degenerates to the base metric. The integration by parts behind Θ-C(1) needs either a Dirichlet or a Neumann boundary condition; on the penteract the latter is the one the kernel limit actually imposes.

(Q4) Derivation. Two derivations meet here. The energy: integrate $\langle\mathcal{L}_\star\phi, \phi\rangle_{p^2}$ by parts, the weight p^2 cancelling against $1/p^2$, to obtain $\frac{1}{2}\int \|\nabla\phi\|^2_{G_5^{-1}} p^2$. The metric: differentiate the embedding $\Phi(c,u) = (c, r(c)u)$, use $u\cdot du = \frac{1}{2}d(\|u\|^2) = 0$ to kill the cross term and $\|u\| = 1$ to normalise the rest, and collect $(\nabla r\cdot dc)^2 = dc^\top \nabla r \nabla r^\top dc$.

(Q5) Retraction through the tower. Replacing ∇ by lattice differences returns the 8S quadratic form $x^\top \mathcal{L}_{8S} x$, and freezing r to a constant collapses ds^2 to the chord $\delta^{(8)}$. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \; 0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta_{\text{ind}}^{(43)})]$ vanishes with $\nu \ge 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{43}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: The Rayleigh quotient bound.

Structure: Identical to entry 27 of field 05 — the ground state energy is the minimum of the Rayleigh quotient. Second-order accuracy means an $O(\epsilon)$ error in the trial state gives $O(\epsilon^2)$ error in the energy, which is why variational methods are so effective.

Reduction: None needed.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T3.

Calculus status: Exact linear; carried to the continuum by Lemma D3, and the input was algebraic, so the differential content is entirely supplied by the transform.

Also in: Quantum Chemistry.

44. Fermi's Golden Rule

Origin (LA edition).

$$\Gamma_{i\to f} = \frac{2\pi}{\hbar}\big|\langle f|\, H'\, |i\rangle\big|^2 \rho(E_f)$$

Input (8S edition).

$$\widehat{W}^{(44)}(k) = \sum_m W(m)\, e^{-2\pi i\langle k,m\rangle/L}, \qquad F_5 = F^{\otimes 5}, \qquad F_5 W F_5^{*} = \mathrm{diag}(\widehat{W}(k))$$

Calculus image.

$$\widehat{W_\sigma * \psi}_{44}(\xi) = \widehat{W_\sigma}(\xi)\,\widehat{\psi_{44}}(\xi), \qquad \widehat{W_\sigma}(\xi) = e^{-\sigma_c^2 \xi^{\top} G_5^{-1} \xi/2}, \qquad \widehat{\partial_a \psi_{44}}(\xi) = i\,\xi_a\,\widehat{\psi_{44}}(\xi)$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T12**, so the calculus image is of class **D12** (Fourier transform and pseudodifferential symbol). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the Fourier transform and pseudodifferential symbol of Lemma D12, instantiated at the point $c_{44} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D12 needs $W_\sigma \in L^1$ for the convolution theorem, which the Gaussian satisfies, and ψ in a space where the transform is defined; $L^1 \cap L^2$ suffices and is standing. Translation invariance of the mask is required for W_σ to act as a convolution at all.

(Q4) Derivation. Apply the convolution theorem, then evaluate the transform of the Gaussian by completing the square in the exponent: $\int e^{-x^{\top}Ax} e^{-i\xi\cdot x} dx = \pi^{5/2} \det(A)^{-1/2} e^{-\xi^{\top} A^{-1}\xi/4}$ with $A = G_5/2\sigma_c^2$. The differentiation rule follows by integrating by parts in the transform integral, the boundary term vanishing by decay. Together these exhibit the coupling as a pseudodifferential operator with Gaussian symbol.

(Q5) Retraction through the tower. On the lattice torus the Fourier transform becomes the five-fold DFT $F_5 = F^{\otimes 5}$ and the symbol becomes the eigenvalue $\widehat{W}(k)$ of the 8S circulant. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \;\, 0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\mathrm{ind}} - \eta_{\mathrm{ind}}^{(44)})]$ vanishes with $\nu \geq 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{44}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: A squared matrix element times a density of states.

Structure: Transition rate is the squared modulus of an off-diagonal matrix element — first-order perturbation theory. The density of states is the spectral density of the final-state Hamiltonian, so the rule is a product of two spectral quantities.

Reduction: The rate is a squared matrix element times a spectral density.

Status: Exact linear (first order).

8S status: Exact linear (first order); preserved under the transform by Theorem C and Lemma T12.

Calculus status: Exact linear (first order); carried to the continuum by Lemma D12, and the input was algebraic, so the differential content is entirely supplied by the transform.

45. Rabi Oscillation

Origin (LA edition).

$$P_e(t) = \frac{\Omega^2}{\Omega^2 + \delta^2} \sin^2\left(\frac{\sqrt{\Omega^2 + \delta^2}}{2} t\right)$$

Input (8S edition).

$$\mathcal{L}_{8S}\,\phi_k^{(45)} = \mu_k^{(45)}\phi_k^{(45)}, \qquad \mathcal{L}_{8S} = K_2 L_W + 2K_3 L_{\hat{H}}, \qquad 0 = \mu_1^{(45)} \le \mu_2^{(45)} \le \cdots \le \mu_N^{(45)}$$

Calculus image.

$$-\frac{1}{2p^2}\nabla\cdot(p^2 G_5^{-1}\nabla\phi_k^{(45)}) = \mu_k^{(45)}\phi_k^{(45)}, \qquad \mu_2^{(45)} = \min_{\substack{\phi\neq 0 \\ \int \phi\, p^2 dc = 0}} \frac{\int_{\mathcal{C}} \|\nabla\phi\|^2_{G_5^{-1}}\, p^2\, dc}{\int_{\mathcal{C}} \phi^2\, p^2\, dc}$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T2**, so the calculus image is of class **D2** (Sturm–Liouville problem and Rayleigh quotient). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the Sturm–Liouville problem and Rayleigh quotient of Lemma D2, instantiated at the point $c_{45} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D2 needs **(R1)**, **(R2)** and the standing normalisation **(N)**. Discreteness of the spectrum uses compactness of $\mathcal{C}$ and the Rellich–Kondrachov embedding $H^1 \hookrightarrow L^2$; without compactness the spectrum could be continuous and μ_2 would not be attained. The min–max formula needs self-adjointness on $L^2(p^2 dc)$, which is Θ-C(2).

(Q4) Derivation. Θ-G supplies the operator by the Gaussian-moment expansion; Θ-C(1) converts it into the bilinear form by the divergence theorem, the boundary term vanishing under the Neumann condition the kernel limit imposes. Imposing the normalisation $\int \phi^2 p^2 = 1$ and taking the first variation of $\mathcal{E}$ produces the eigenvalue equation with μ as the Lagrange multiplier; Courant–Fischer then gives the displayed variational characterisation of μ_2 on the complement of the constants.

(Q5) Retraction through the tower. Discretising the Rayleigh quotient on $\mathcal{C}_M$ returns $x^\top \mathcal{L}_{8S} x / x^\top x$, whose minimiser over mean-zero x is the 8S μ_2; the gradient becomes a difference quotient with $O(\ell^2)$ error. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \;\, 0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta_{\text{ind}}^{(45)})]$ vanishes with $\nu \ge 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{45}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: Exact solution of a 2×2 Hermitian system.

Structure: The two-level Hamiltonian $\frac{\hbar}{2}\left(\begin{smallmatrix} -\delta & \Omega \\ \Omega & \delta \end{smallmatrix}\right)$ has eigenvalue splitting $\sqrt{\Omega^2 + \delta^2}$, and that splitting is the oscillation frequency. On resonance ($\delta = 0$) the transfer is complete because the eigenvectors are equal superpositions — an avoided crossing at its symmetric point.

Reduction: None needed.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T2.

Calculus status: Exact linear; carried to the continuum by Lemma D2, and the input was algebraic, so the differential content is entirely supplied by the transform.

46. Adiabatic Theorem

Origin (LA edition).

$$\text{slow change} \implies \text{the system stays in the instantaneous eigenvector}$$

Input (8S edition).

$$\mathcal{L}_{8S}\,\phi_k^{(46)} = \mu_k^{(46)}\phi_k^{(46)}, \qquad \mathcal{L}_{8S} = K_2 L_W + 2K_3 L_{\hat{H}}, \qquad 0 = \mu_1^{(46)} \le \mu_2^{(46)} \le \cdots \le \mu_N^{(46)}$$

Calculus image.

$$-\frac{1}{2p^2}\nabla\cdot(p^2 G_5^{-1}\nabla\phi_k^{(46)}) = \mu_k^{(46)}\phi_k^{(46)}, \qquad \mu_2^{(46)} = \min_{\substack{\phi\neq 0 \\ \int \phi\, p^2 dc = 0}} \frac{\int_{\mathcal{C}} \|\nabla\phi\|^2_{G_5^{-1}}\, p^2\, dc}{\int_{\mathcal{C}} \phi^2\, p^2\, dc}$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T2**, so the calculus image is of class **D2** (Sturm–Liouville problem and Rayleigh quotient). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the Sturm–Liouville problem and Rayleigh quotient of Lemma D2, instantiated at the point $c_{46} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D2 needs **(R1)**, **(R2)** and the standing normalisation **(N)**. Discreteness of the spectrum uses compactness of $\mathcal{C}$ and the Rellich–Kondrachov embedding $H^1 \hookrightarrow L^2$; without compactness the spectrum could be continuous and μ_2 would not be attained. The min–max formula needs self-adjointness on $L^2(p^2 dc)$, which is Θ-C(2).

(Q4) Derivation. Θ-G supplies the operator by the Gaussian-moment expansion; Θ-C(1) converts it into the bilinear form by the divergence theorem, the boundary term vanishing under the Neumann condition the kernel limit imposes. Imposing the normalisation $\int \phi^2 p^2 = 1$ and taking the first variation of $\mathcal{E}$ produces the eigenvalue equation with μ as the Lagrange multiplier; Courant–Fischer then gives the displayed variational characterisation of μ_2 on the complement of the constants.

(Q5) Retraction through the tower. Discretising the Rayleigh quotient on $\mathcal{C}_M$ returns $x^\top \mathcal{L}_{8S} x / x^\top x$, whose minimiser over mean-zero x is the 8S μ_2; the gradient becomes a difference quotient with $O(\ell^2)$ error. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \;\, 0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta_{\text{ind}}^{(46)})]$ vanishes with $\nu \ge 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{46}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: Continuous tracking of an eigenvector as the matrix varies.

Structure: Valid when the rate of change is small compared with the squared gap: $|\langle m|\dot{H}|n\rangle| \ll |E_n - E_m|^2/\hbar$. Small gaps break adiabaticity — Landau–Zener transitions occur at avoided crossings, which is exactly eigenvector ill-conditioning.

Reduction: The theorem is eigenvector continuity under slow variation.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T2.

Calculus status: Exact linear; carried to the continuum by Lemma D2, and the input was algebraic, so the differential content is entirely supplied by the transform.

47. Born Approximation

Origin (LA edition).

$$f(\theta) = -\frac{m}{2\pi\hbar^2}\int e^{-i\mathbf{q}\cdot\mathbf{r}} V(\mathbf{r})\, d^3 r$$

Input (8S edition).

$$\widehat{W}^{(47)}(k) = \sum_m W(m)\, e^{-2\pi i\langle k,m\rangle/L}, \qquad F_5 = F^{\otimes 5}, \qquad F_5 W F_5^{*} = \mathrm{diag}(\widehat{W}(k))$$

Calculus image.

$$\widehat{W_\sigma * \psi}_{47}(\xi) = \widehat{W_\sigma}(\xi)\, \widehat{\psi_{47}}(\xi), \qquad \widehat{W_\sigma}(\xi) = e^{-\sigma_c^2 \xi^\top G_5^{-1} \xi/2}, \qquad \widehat{\partial_a \psi_{47}}(\xi) = i\,\xi_a\, \widehat{\psi_{47}}(\xi)$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T12**, so the calculus image is of class **D12** (Fourier transform and pseudodifferential symbol). The originating equation is first order — it already carries a derivative or an integral. The input already carries a first-order differential or integral operator. On that part the transform acts as the identity — a derivative is already a calculus object — and only the coupling and the gate are taken to the continuum. The regularity demanded of the input is correspondingly one order lower.

(Q2) Image. The displayed relation is the Fourier transform and pseudodifferential symbol of Lemma D12, instantiated at the point $c_{47} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D12 needs $W_\sigma \in L^1$ for the convolution theorem, which the Gaussian satisfies, and ψ in a space where the transform is defined; $L^1 \cap L^2$ suffices and is standing. Translation invariance of the mask is required for W_σ to act as a convolution at all.

(Q4) Derivation. Apply the convolution theorem, then evaluate the transform of the Gaussian by completing the square in the exponent: $\int e^{-x^\top A x} e^{-i\xi\cdot x} dx = \pi^{5/2} \det(A)^{-1/2} e^{-\xi^\top A^{-1}\xi/4}$ with $A = G_5/2\sigma_c^2$. The differentiation rule follows by integrating by parts in the transform integral, the boundary term vanishing by decay. Together these exhibit the coupling as a pseudodifferential operator with Gaussian symbol.

(Q5) Retraction through the tower. On the lattice torus the Fourier transform becomes the five-fold DFT $F_5 = F^{\otimes 5}$ and the symbol becomes the eigenvalue $\widehat{W}(k)$ of the 8S circulant. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \;\, 0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\mathrm{ind}} - \eta_{\mathrm{ind}}^{(47)})]$ vanishes with $\nu \geq 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{47}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: The scattering amplitude as the Fourier transform of the potential.

Structure: First term of the Born (Neumann) series $T = V + VG_0V + \cdots$ — a resolvent expansion, convergent when the potential is weak. The amplitude is a linear transform of the potential, which is why scattering experiments invert to structure.

Reduction: The amplitude is a Fourier transform of the potential.

Status: Exact linear (first order).

8S status: Exact linear (first order); preserved under the transform by Theorem C and Lemma T12.

Calculus status: Exact linear (first order); carried to the continuum by Lemma D12, and the input already carried a first-order operator, which the transform leaves in place.

Also in: Nuclear Physics.

48. Partial Wave Expansion

Origin (LA edition).

$$f(\theta) = \frac{1}{k} \sum_\ell (2\ell + 1) e^{i\delta_\ell} \sin\delta_\ell\, P_\ell(\cos\theta)$$

Input (8S edition).

$$y_{48}(\tau,u)=\Pi(\tau)\Big[(\Gamma W\Gamma)\otimes \mathsf{B}(\mathsf{A}_{48})\Big]z(u),\qquad z_{48}(u)=\begin{bmatrix}c_{48}\\ r_{48}u\end{bmatrix}$$

Calculus image.

$$(\mathcal{K}_\sigma\psi_{48})(c)=\int_{\mathcal{C}}\gamma(c)\,W_\sigma(c,c')\,\gamma(c')\,\mathsf{A}_{48}\,\psi_{48}(c')\,p(c')\,dc',\qquad \partial_\tau\psi_{48}=-\mathcal{L}_\star\psi_{48},\qquad \mathcal{L}_\star=-\frac{1}{2p^2}\nabla\cdot(p^2G_5^{-1}\nabla\,\cdot\,)$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T1**, so the calculus image is of class **D1** (Fredholm integral operator and its generator). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the Fredholm integral operator and its generator of Lemma D1, instantiated at the point $c_{48}\in\mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D1 needs the kernel continuous and square-integrable on $\mathcal{C}\times\mathcal{C}$, which the Gaussian form supplies unconditionally, so $\mathcal{K}_\sigma$ is Hilbert–Schmidt and hence compact. The generator statement needs **(R1)** and **(R2)**: C^4 regularity for the $O(\sigma^2)$ rate of Θ-G, and interior evaluation, since within $O(\sigma)$ of $\partial\mathcal{C}$ the expansion picks up the Neumann boundary term of Remark Θ-G″.

(Q4) Derivation. The 8S image is a Kronecker product $(\Gamma W\Gamma)\otimes\mathsf{B}(\mathsf{A})$ acting by a sum over nodes. Definition 2.2 replaces $\sum_j(\cdot)_j$ by $\int_{\mathcal{C}}(\cdot)\,p\,dc'$, turning the matrix into the displayed Fredholm operator of the second kind. Its bandwidth limit is computed by Θ-G: Taylor-expand ψ and p to second order about c, integrate against the Gaussian, use the vanishing of the odd moments and $\int w\,uu^{\top}du=\sigma^2\kappa\,G_5^{-1}$, and divide by the degree. What survives at order σ^2 is $-\mathcal{L}_\star$.

(Q5) Retraction through the tower. Sampling the integral operator on $\mathcal{C}_M$ by the Riemann rule of Θ-J(1) returns the finite sum $(\Gamma W\Gamma)\otimes\mathsf{B}(\mathsf{A}_{48})$ with $O(\ell^2)$ quadrature error. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma\to I$, $W\to I$, $\mathcal{R}\to 0$, $K_2,K_3,J_2,J_3\to 0$ and $\Pi\to[\,I_3\;0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S}-\nu(\varepsilon_{\mathrm{ind}}-\eta_{\mathrm{ind}}^{(48)})]$ vanishes with $\nu\ge 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{48}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: Decomposition in the angular momentum eigenbasis.

Structure: Rotational symmetry block-diagonalises the scattering operator by ℓ, so the S-matrix is diagonal with entries $e^{2i\delta_\ell}$. Unitarity is $|S_\ell|=1$ for elastic scattering, and each phase shift is one diagonal entry.

Reduction: Symmetry diagonalises the scattering matrix.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T1.

Calculus status: Exact linear; carried to the continuum by Lemma D1, and the input was algebraic, so the differential content is entirely supplied by the transform.

49. Optical Theorem

Origin (LA edition).

$$\sigma_{\mathrm{tot}}=\frac{4\pi}{k}\,\mathrm{Im}\,f(0)$$

Input (8S edition).

$$y_{49}(\tau,u)=\Pi(\tau)\Big[(\Gamma W\Gamma)\otimes \mathsf{B}(\mathsf{A}_{49})\Big]z(u),\qquad z_{49}(u)=\begin{bmatrix}c_{49}\\ r_{49}u\end{bmatrix}$$

Calculus image.

$$(\mathcal{K}_\sigma\psi_{49})(c) = \int_{\mathcal{C}} \gamma(c)\, W_\sigma(c,c')\, \gamma(c')\, \mathsf{A}_{49}\, \psi_{49}(c')\, p(c')\, dc', \qquad \partial_\tau \psi_{49} = -\mathcal{L}_\star \psi_{49}, \qquad \mathcal{L}_\star = -\frac{1}{2p^2}\nabla\cdot(p^2 G_5^{-1}\nabla\,\cdot\,)$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T1**, so the calculus image is of class **D1** (Fredholm integral operator and its generator). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the Fredholm integral operator and its generator of Lemma D1, instantiated at the point $c_{49} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D1 needs the kernel continuous and square-integrable on $\mathcal{C}\times\mathcal{C}$, which the Gaussian form supplies unconditionally, so $\mathcal{K}_\sigma$ is Hilbert–Schmidt and hence compact. The generator statement needs **(R1)** and **(R2)**: C^4 regularity for the $O(\sigma^2)$ rate of Θ-G, and interior evaluation, since within $O(\sigma)$ of $\partial\mathcal{C}$ the expansion picks up the Neumann boundary term of Remark Θ-G″.

(Q4) Derivation. The 8S image is a Kronecker product $(\Gamma W\Gamma)\otimes \mathsf{B}(\mathsf{A})$ acting by a sum over nodes. Definition 2.2 replaces $\sum_j (\cdot)_j$ by $\int_{\mathcal{C}}(\cdot)\,p\,dc'$, turning the matrix into the displayed Fredholm operator of the second kind. Its bandwidth limit is computed by Θ-G: Taylor-expand ψ and p to second order about c, integrate against the Gaussian, use the vanishing of the odd moments and $\int w\, uu^\top du = \sigma^2\kappa\, G_5^{-1}$, and divide by the degree. What survives at order σ^2 is $-\mathcal{L}_\star$.

(Q5) Retraction through the tower. Sampling the integral operator on $\mathcal{C}_M$ by the Riemann rule of Θ-J(1) returns the finite sum $(\Gamma W\Gamma)\otimes\mathsf{B}(\mathsf{A}_{49})$ with $O(\ell^2)$ quadrature error. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma\to I$, $W\to I$, $\mathcal{R}\to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\,I_3\ 0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta_{\text{ind}}^{(49)})]$ vanishes with $\nu \geq 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{49}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: A consequence of unitarity of the S-matrix.

Structure: $S^\dagger S = I$ relates the forward amplitude's imaginary part to the total cross section — probability removed from the forward beam must reappear elsewhere. It is unitarity written out for one matrix element.

Reduction: The theorem is S-matrix unitarity.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T1.

Calculus status: Exact linear; carried to the continuum by Lemma D1, and the input was algebraic, so the differential content is entirely supplied by the transform.

50. Scattering Cross Section

Origin (LA edition).

$$\frac{d\sigma}{d\Omega} = \big|f(\theta)\big|^2$$

Input (8S edition).

$$E_{50}^{8S}(z) = z^\top \mathcal{G}_8 z\Big|_{\text{node }50}, \qquad \big\|z_{50}(u) - z_j(u)\big\|_{\mathcal{G}_8} = \delta_{50j}^{(8)} = \sqrt{(c_{50}-c_j)^\top G_5 (c_{50}-c_j) + (r_{50}-r_j)^2}$$

Calculus image.

$$\mathcal{E}_{50}[\phi] = \tfrac{1}{2}\int_{\mathcal{C}} \|\nabla\phi\|^2_{G_5^{-1}}\, p^2\, dc, \qquad \delta\mathcal{E}_{50} = 0 \iff \mathcal{L}_\star\phi = 0, \qquad ds^2_{50} = dc^\top(G_5 + \nabla r\,\nabla r^\top)dc + r^2_{50}\, d\Omega^2_3$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T3**, so the calculus image is of class **D3** (Dirichlet energy and the cone metric). The originating equation is first order — it already carries a derivative or an integral. The input already carries a first-order differential or integral operator. On that part the transform acts as the identity — a derivative is already a calculus object — and only the coupling and the gate are taken to the continuum. The regularity demanded of the input is correspondingly one order lower.

(Q2) Image. The displayed relation is the Dirichlet energy and the cone metric of Lemma D3, instantiated at the point $c_{50} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D3 needs **(R3)**, $r \in C^1$ with $r > 0$, for the bundle to be a smooth manifold; at $r = 0$ the fibre collapses and ds^2 degenerates to the base metric. The integration by parts behind Θ-C(1) needs either a Dirichlet or a Neumann boundary condition; on the penteract the latter is the one the kernel limit actually imposes.

(Q4) Derivation. Two derivations meet here. The energy: integrate $\langle\mathcal{L}_\star\phi, \phi\rangle_{p^2}$ by parts, the weight p^2 cancelling against $1/p^2$, to obtain $\frac{1}{2}\int \|\nabla\phi\|^2_{G_5^{-1}} p^2$. The metric: differentiate the embedding $\Phi(c, u) = (c, r(c)u)$, use $u\cdot du = \frac{1}{2}d(\|u\|^2) = 0$ to kill the cross term and $\|u\| = 1$ to normalise the rest, and collect $(\nabla r\cdot dc)^2 = dc^\top\nabla r\nabla r^\top dc$.

(Q5) Retraction through the tower. Replacing ∇ by lattice differences returns the 8S quadratic form $x^\top\mathcal{L}_{8S}x$, and freezing r to a constant collapses ds^2 to the chord $\delta^{(8)}$. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\,I_3\ 0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta^{(50)}_{\text{ind}})]$ vanishes with $\nu \geq 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{50}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: The squared modulus of a matrix element.

Structure: Amplitudes add; cross sections do not — interference between scattering channels is the cross term. This is the Born rule applied to scattering, so the same quadratic-form structure recurs.

Reduction: The cross section is a squared amplitude.

Status: Exact linear in the amplitude.

8S status: Exact linear in the amplitude; preserved under the transform by Theorem C and Lemma T3.

Calculus status: Exact linear in the amplitude; carried to the continuum by Lemma D3, and the input already carried a first-order operator, which the transform leaves in place.

Identical Particles and Quantum Information

51. Symmetrisation Postulate

Origin (LA edition).

$$\hat{P}_{12}\,|\psi\rangle = \pm\,|\psi\rangle$$

Input (8S edition).

$$\mathcal{L}_{8S}\,\phi^{(51)}_k = \mu^{(51)}_k\phi^{(51)}_k, \qquad \mathcal{L}_{8S} = K_2L_W + 2K_3L_{\hat{H}}, \qquad 0 = \mu^{(51)}_1 \leq \mu^{(51)}_2 \leq \cdots \leq \mu^{(51)}_N$$

Calculus image.

$$-\frac{1}{2p^2}\nabla\cdot(p^2 G_5^{-1}\nabla\phi_k^{(51)}) = \mu_k^{(51)}\phi_k^{(51)}, \qquad \mu_2^{(51)} = \min_{\substack{\phi\neq 0 \\ \int \phi\, p^2 dc = 0}} \frac{\int_{\mathcal{C}} \|\nabla\phi\|^2_{G_5^{-1}}\, p^2\, dc}{\int_{\mathcal{C}} \phi^2\, p^2\, dc}$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T2**, so the calculus image is of class **D2** (Sturm–Liouville problem and Rayleigh quotient). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the Sturm–Liouville problem and Rayleigh quotient of Lemma D2, instantiated at the point $c_{51} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D2 needs **(R1)**, **(R2)** and the standing normalisation **(N)**. Discreteness of the spectrum uses compactness of $\mathcal{C}$ and the Rellich–Kondrachov embedding $H^1 \hookrightarrow L^2$; without compactness the spectrum could be continuous and μ_2 would not be attained. The min–max formula needs self-adjointness on $L^2(p^2 dc)$, which is Θ-C(2).

(Q4) Derivation. Θ-G supplies the operator by the Gaussian-moment expansion; Θ-C(1) converts it into the bilinear form by the divergence theorem, the boundary term vanishing under the Neumann condition the kernel limit imposes. Imposing the normalisation $\int \phi^2 p^2 = 1$ and taking the first variation of $\mathcal{E}$ produces the eigenvalue equation with μ as the Lagrange multiplier; Courant–Fischer then gives the displayed variational characterisation of μ_2 on the complement of the constants.

(Q5) Retraction through the tower. Discretising the Rayleigh quotient on $\mathcal{C}_M$ returns $x^\top \mathcal{L}_{8S} x / x^\top x$, whose minimiser over mean-zero x is the 8S μ_2; the gradient becomes a difference quotient with $O(\ell^2)$ error. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \;\, 0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta_{\text{ind}}^{(51)})]$ vanishes with $\nu \geq 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{51}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: Eigenvectors of the exchange operator with eigenvalue ± 1.

Structure: The exchange operator satisfies $\hat{P}^2 = I$, so its eigenvalues are ± 1 and the state space splits into symmetric and antisymmetric subspaces. Bosons and fermions are the two eigenspaces of one involution.

Reduction: Statistics is an eigenvalue of the exchange operator.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T2.

Calculus status: Exact linear; carried to the continuum by Lemma D2, and the input was algebraic, so the differential content is entirely supplied by the transform.

52. Pauli Exclusion Principle

Origin (LA edition).

$$|\psi\rangle = 0 \text{ if two fermions share a state}$$

Input (8S edition).

$$y_{52}(\tau, u) = \Pi(\tau)\Big[(\Gamma W \Gamma) \otimes \mathsf{B}(\mathsf{A}_{52})\Big] z(u), \qquad z_{52}(u) = \begin{bmatrix} c_{52} \\ r_{52} u \end{bmatrix}$$

Calculus image.

$$(\mathcal{K}_\sigma\psi_{52})(c) = \int_{\mathcal{C}} \gamma(c)\, W_\sigma(c,c')\, \gamma(c')\, \mathsf{A}_{52}\, \psi_{52}(c')\, p(c')\, dc', \qquad \partial_\tau \psi_{52} = -\mathcal{L}_\star \psi_{52}, \qquad \mathcal{L}_\star = -\frac{1}{2p^2}\nabla\cdot(p^2 G_5^{-1}\nabla\,\cdot\,)$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T1**, so the calculus image is of class **D1** (Fredholm integral operator and its generator). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the Fredholm integral operator and its generator of Lemma D1, instantiated at the point $c_{52} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D1 needs the kernel continuous and square-integrable on $\mathcal{C}\times\mathcal{C}$, which the Gaussian form supplies unconditionally, so $\mathcal{K}_\sigma$ is Hilbert–Schmidt and hence compact. The generator statement needs **(R1)** and **(R2)**: C^4 regularity for the $O(\sigma^2)$ rate of Θ-G, and interior evaluation, since within $O(\sigma)$ of $\partial\mathcal{C}$ the expansion picks up the Neumann boundary term of Remark Θ-G″.

(Q4) Derivation. The 8S image is a Kronecker product $(\Gamma W\Gamma)\otimes \mathsf{B}(\mathsf{A})$ acting by a sum over nodes. Definition 2.2 replaces $\sum_j(\cdot)_j$ by $\int_{\mathcal{C}}(\cdot)\,p\,dc'$, turning the matrix into the displayed Fredholm operator of the second kind. Its bandwidth limit is computed by Θ-G: Taylor-expand ψ and p to second order about c, integrate against the Gaussian, use the vanishing of the odd moments and $\int w\,uu^\top du = \sigma^2\kappa\, G_5^{-1}$, and divide by the degree. What survives at order σ^2 is $-\mathcal{L}_\star$.

(Q5) Retraction through the tower. Sampling the integral operator on $\mathcal{C}_M$ by the Riemann rule of Θ-J(1) returns the finite sum $(\Gamma W\Gamma)\otimes\mathsf{B}(\mathsf{A}_{52})$ with $O(\ell^2)$ quadrature error. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma\to I$, $W\to I$, $\mathcal{R}\to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\,I_3\ 0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta^{(52)}_{\text{ind}})]$ vanishes with $\nu \geq 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{52}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: Vanishing of an antisymmetrised product.

Structure: The antisymmetriser applied to a repeated state gives zero, because a determinant with two equal columns vanishes. Exclusion is the vanishing of a determinant — the sharpest possible linear-algebraic statement of it.

Reduction: Exclusion is a vanishing determinant.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T1.

Calculus status: Exact linear; carried to the continuum by Lemma D1, and the input was algebraic, so the differential content is entirely supplied by the transform.

53. Slater Determinant

Origin (LA edition).

$$\Psi = \frac{1}{\sqrt{N!}}\det[\phi_i(x_j)]$$

Input (8S edition).

$$\max_{n_{53}\in\mathbb{Z}^5} \Delta_{8S} = \text{Score}(\mathcal{M}_8) - \text{Score}(\mathcal{M}_7) \quad \text{s.t.} \quad \|n_{53}\|_1 \leq M,\ \ \eta^{(53)}_{\text{ind}} \geq \varepsilon_{\text{ind}}$$

Calculus image.

$$\delta\Big[\Delta_{8S} - \lambda(\|n_{53}\|_1 - M) - \nu(\varepsilon_{\text{ind}} - \eta_{\text{ind}}^{(53)})\Big] = 0, \qquad \lambda, \nu \geq 0, \qquad \lambda(\|n_{53}\|_1 - M) = \nu(\varepsilon_{\text{ind}} - \eta_{\text{ind}}^{(53)}) = 0$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T11**, so the calculus image is of class **D11** (Euler–Lagrange equation and KKT conditions). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the Euler–Lagrange equation and KKT conditions of Lemma D11, instantiated at the point $c_{53} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D11 needs the feasible set convex — the ℓ_1 ball is — and the objective concave in the relaxed variables, which is the standing assumption on Δ_{8S}. Under those two, KKT is necessary and sufficient; without concavity it is necessary only.

(Q4) Derivation. Relax the integrality of n and form the Lagrangian of the 8S programme. Stationarity in the relaxed variable is the Euler–Lagrange condition; the sign conditions $\lambda, \nu \geq 0$ come from the inequality constraints and complementary slackness records which of them binds. Concavity of the objective on a convex feasible set upgrades necessity to sufficiency.

(Q5) Retraction through the tower. On the lattice the variation becomes a finite difference and the KKT system becomes the optimality conditions of the 8S linear programme over the cross-polytope. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \; 0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta_{\text{ind}}^{(53)})]$ vanishes with $\nu \geq 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{53}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: An antisymmetrised product — literally a determinant.

Structure: Spans the antisymmetric subspace of $\bigotimes^N \mathcal{H}$, of dimension $\binom{d}{N}$. Invariant under orthogonal transformations of the occupied orbitals, so only the occupied *subspace* matters — Hartree–Fock optimises a subspace, not a basis.

Reduction: The state is a determinant.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T11.

Calculus status: Exact linear; carried to the continuum by Lemma D11, and the input was algebraic, so the differential content is entirely supplied by the transform.

Also in: Quantum Chemistry.

54. Exchange Interaction

Origin (LA edition).

$$E_{\text{exch}} = -\int \phi_a^*(1)\phi_b^*(2)\frac{e^2}{r_{12}}\phi_b(1)\phi_a(2)$$

Input (8S edition).

$$y_{54}(\tau, u) = \Pi(\tau)\Big[(\Gamma W \Gamma) \otimes \mathsf{B}(\mathsf{A}_{54})\Big] z(u), \qquad z_{54}(u) = \begin{bmatrix} c_{54} \\ r_{54} u \end{bmatrix}$$

Calculus image.

$$(\mathcal{K}_\sigma \psi_{54})(c) = \int_{\mathcal{C}} \gamma(c)\, W_\sigma(c, c')\, \gamma(c')\, \mathsf{A}_{54}\, \psi_{54}(c')\, p(c')\, dc', \qquad \partial_\tau \psi_{54} = -\mathcal{L}_\star \psi_{54}, \qquad \mathcal{L}_\star = -\frac{1}{2p^2}\nabla \cdot (p^2 G_5^{-1} \nabla\, \cdot\,)$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T1**, so the calculus image is of class **D1** (Fredholm integral operator and its generator). The originating equation is first order — it already carries a derivative or an integral. The input already carries a first-order differential or integral operator. On that part the transform acts as the identity — a derivative is already a calculus object — and only the coupling and the gate are taken to the continuum. The regularity demanded of the input is correspondingly one order lower.

(Q2) Image. The displayed relation is the Fredholm integral operator and its generator of Lemma D1, instantiated at the point $c_{54} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D1 needs the kernel continuous and square-integrable on $\mathcal{C} \times \mathcal{C}$, which the Gaussian form supplies unconditionally, so $\mathcal{K}_\sigma$ is Hilbert–Schmidt and hence compact. The generator statement needs **(R1)** and **(R2)**: C^4 regularity for the $O(\sigma^2)$ rate of Θ-G, and interior evaluation, since within $O(\sigma)$ of $\partial\mathcal{C}$ the expansion picks up the Neumann boundary term of Remark Θ-G″.

(Q4) Derivation. The 8S image is a Kronecker product $(\Gamma W \Gamma) \otimes \mathsf{B}(\mathsf{A})$ acting by a sum over nodes. Definition 2.2 replaces $\sum_j (\cdot)_j$ by $\int_{\mathcal{C}} (\cdot)\, p\, dc'$, turning the matrix into the displayed Fredholm operator of the second kind. Its bandwidth limit is computed by Θ-G: Taylor-expand ψ and p to second order about c, integrate against the Gaussian, use the vanishing of the odd moments and $\int w\, uu^\top du = \sigma^2 \kappa\, G_5^{-1}$, and divide by the degree. What survives at order σ^2 is $-\mathcal{L}_\star$.

(Q5) Retraction through the tower. Sampling the integral operator on $\mathcal{C}_M$ by the Riemann rule of Θ-J(1) returns the finite sum $(\Gamma W \Gamma) \otimes \mathsf{B}(\mathsf{A}_{54})$ with $O(\ell^2)$ quadrature error. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \; 0 \,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta_{\text{ind}}^{(54)})]$ vanishes with $\nu \geq 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{54}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: An off-diagonal matrix element arising from antisymmetry.

Structure: No classical analogue — it comes entirely from the determinant structure of the state. It is what produces ferromagnetism and Hund's rules, both consequences of a sign in a permutation.

Reduction: The interaction is an antisymmetry-induced matrix element.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T1.

Calculus status: Exact linear; carried to the continuum by Lemma D1, and the input already carried a first-order operator, which the transform leaves in place.

55. Entangled State (Bell State)

Origin (LA edition).

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

Input (8S edition).

$$\mathcal{L}_{8S}\, \phi_k^{(55)} = \mu_k^{(55)} \phi_k^{(55)}, \qquad \mathcal{L}_{8S} = K_2 L_W + 2K_3 L_{\hat{H}}, \qquad 0 = \mu_1^{(55)} \leq \mu_2^{(55)} \leq \cdots \leq \mu_N^{(55)}$$

Calculus image.

$$-\frac{1}{2p^2} \nabla \cdot (p^2 G_5^{-1} \nabla \phi_k^{(55)}) = \mu_k^{(55)} \phi_k^{(55)}, \qquad \mu_2^{(55)} = \min_{\substack{\phi \neq 0 \\ \int \phi\, p^2 dc = 0}} \frac{\int_{\mathcal{C}} \|\nabla \phi\|^2_{G_5^{-1}}\, p^2\, dc}{\int_{\mathcal{C}} \phi^2\, p^2\, dc}$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T2**, so the calculus image is of class **D2** (Sturm–Liouville problem and Rayleigh quotient). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the Sturm–Liouville problem and Rayleigh quotient of Lemma D2, instantiated at the point $c_{55} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D2 needs **(R1)**, **(R2)** and the standing normalisation **(N)**. Discreteness of the spectrum uses compactness of $\mathcal{C}$ and the Rellich–Kondrachov embedding $H^1 \hookrightarrow L^2$; without compactness the spectrum could be continuous and μ_2 would not be attained. The min–max formula needs self-adjointness on $L^2(p^2 dc)$, which is Θ-C(2).

(Q4) Derivation. Θ-G supplies the operator by the Gaussian-moment expansion; Θ-C(1) converts it into the bilinear form by the divergence theorem, the boundary term vanishing under the Neumann condition the kernel limit imposes. Imposing the normalisation $\int \phi^2 p^2 = 1$ and taking the first variation of $\mathcal{E}$ produces the eigenvalue equation with μ as the Lagrange multiplier; Courant–Fischer then gives the displayed variational characterisation of μ_2 on the complement of the constants.

(Q5) Retraction through the tower. Discretising the Rayleigh quotient on $\mathcal{C}_M$ returns $x^\top \mathcal{L}_{8S} x / x^\top x$, whose minimiser over mean-zero x is the 8S μ_2; the gradient becomes a difference quotient with $O(\ell^2)$ error. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \;\, 0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta_{\text{ind}}^{(55)})]$ vanishes with $\nu \geq 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{55}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: A rank-two matrix in the bipartite coefficient basis.

Structure: Writing the state as a coefficient matrix C, product states are exactly $\operatorname{rank} C = 1$ and entangled states have higher rank. The Schmidt coefficients are the singular values of C, and entanglement entropy is the von Neumann entropy of σ_i^2. Entanglement is a rank statement, and the SVD is its complete characterisation.

Reduction: Entanglement is matrix rank; Schmidt decomposition is the SVD.

Status: Exact linear. Possibly the cleanest single insight in the edition.

8S status: Exact linear. Possibly the cleanest single insight in the edition; preserved under the transform by Theorem C and Lemma T2.

Calculus status: Exact linear. Possibly the cleanest single insight in the edition; carried to the continuum by Lemma D2, and the input was algebraic, so the differential content is entirely supplied by the transform.

56. CHSH Bell Inequality

Origin (LA edition).

$$\big|\langle A_1 B_1\rangle + \langle A_1 B_2\rangle + \langle A_2 B_1\rangle - \langle A_2 B_2\rangle\big| \leq 2$$

Input (8S edition).

$$y_{56}(\tau, u) = \Pi(\tau)\Big[(\Gamma W \Gamma) \otimes \mathsf{B}(\mathsf{A}_{56})\Big] z(u), \qquad z_{56}(u) = \begin{bmatrix} c_{56} \\ r_{56} u \end{bmatrix}$$

Calculus image.

$$(\mathcal{K}_\sigma \psi_{56})(c) = \int_{\mathcal{C}} \gamma(c)\, W_\sigma(c, c')\, \gamma(c')\, \mathsf{A}_{56}\, \psi_{56}(c')\, p(c')\, dc', \qquad \partial_\tau \psi_{56} = -\mathcal{L}_\star \psi_{56}, \qquad \mathcal{L}_\star = -\frac{1}{2p^2} \nabla \cdot (p^2 G_5^{-1} \nabla\, \cdot\,)$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T1**, so the calculus image is of class **D1** (Fredholm integral operator and its generator). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the Fredholm integral operator and its generator of Lemma D1, instantiated at the point $c_{56} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D1 needs the kernel continuous and square-integrable on $\mathcal{C} \times \mathcal{C}$, which the Gaussian form supplies unconditionally, so $\mathcal{K}_\sigma$ is Hilbert–Schmidt and hence compact. The generator statement needs **(R1)** and **(R2)**: C^4 regularity for the $O(\sigma^2)$ rate of Θ-G, and interior evaluation, since within $O(\sigma)$ of $\partial\mathcal{C}$ the expansion picks up the Neumann boundary term of Remark Θ-G″.

(Q4) Derivation. The 8S image is a Kronecker product $(\Gamma W \Gamma) \otimes \mathsf{B}(\mathsf{A})$ acting by a sum over nodes. Definition 2.2 replaces $\sum_j (\cdot)_j$ by $\int_{\mathcal{C}} (\cdot)\, p\, dc'$, turning the matrix into the displayed Fredholm operator of the second kind. Its bandwidth limit is computed by Θ-G: Taylor-expand ψ and p to second order about c, integrate against the Gaussian, use the vanishing of the odd moments and $\int w\, uu^\top du = \sigma^2 \kappa\, G_5^{-1}$, and divide by the degree. What survives at order σ^2 is $-\mathcal{L}_\star$.

(Q5) Retraction through the tower. Sampling the integral operator on $\mathcal{C}_M$ by the Riemann rule of Θ-J(1) returns the finite sum $(\Gamma W \Gamma) \otimes \mathsf{B}(\mathsf{A}_{56})$ with $O(\ell^2)$ quadrature error. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \; 0 \,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta_{\text{ind}}^{(56)})]$ vanishes with $\nu \geq 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{56}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: A bound on a bilinear form over local hidden variable models.

Structure: Classical models are constrained to a polytope with the CHSH facets; quantum mechanics reaches $2\sqrt{2}$ (Tsirelson), the maximum over the larger set of quantum correlations. The quantum bound comes from an SDP relaxation, so the gap between classical and quantum is a polytope-versus-spectrahedron gap.

Reduction: The bound separates a polytope from a spectrahedron.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T1.

Calculus status: Exact linear; carried to the continuum by Lemma D1, and the input was algebraic, so the differential content is entirely supplied by the transform.

57. No-Cloning Theorem

Origin (LA edition).

$$\nexists\, U : U\,|\psi\rangle\,|0\rangle = |\psi\rangle\,|\psi\rangle \;\; \forall\, |\psi\rangle$$

Input (8S edition).

$$y_{57}(\tau, u) = \Pi(\tau)\Big[(\Gamma W \Gamma) \otimes \mathsf{B}(\mathsf{A}_{57})\Big] z(u), \qquad z_{57}(u) = \begin{bmatrix} c_{57} \\ r_{57} u \end{bmatrix}$$

Calculus image.

$$(\mathcal{K}_\sigma \psi_{57})(c) = \int_{\mathcal{C}} \gamma(c)\, W_\sigma(c, c')\, \gamma(c')\, \mathsf{A}_{57}\, \psi_{57}(c')\, p(c')\, dc', \qquad \partial_\tau \psi_{57} = -\mathcal{L}_\star \psi_{57}, \qquad \mathcal{L}_\star = -\frac{1}{2p^2} \nabla \cdot (p^2 G_5^{-1} \nabla\, \cdot\,)$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T1**, so the calculus image is of class **D1** (Fredholm integral operator and its generator). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the Fredholm integral operator and its generator of Lemma D1, instantiated at the point $c_{57} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D1 needs the kernel continuous and square-integrable on $\mathcal{C} \times \mathcal{C}$, which the Gaussian form supplies unconditionally, so $\mathcal{K}_\sigma$ is Hilbert–Schmidt and hence compact. The generator statement needs **(R1)** and **(R2)**: C^4 regularity for the $O(\sigma^2)$ rate of Θ-G, and interior evaluation, since within $O(\sigma)$ of $\partial\mathcal{C}$ the expansion picks up the Neumann boundary term of Remark Θ-G″.

(Q4) Derivation. The 8S image is a Kronecker product $(\Gamma W \Gamma) \otimes \mathsf{B}(\mathsf{A})$ acting by a sum over nodes. Definition 2.2 replaces $\sum_j (\cdot)_j$ by $\int_{\mathcal{C}} (\cdot)\, p\, dc'$, turning the matrix into the displayed Fredholm operator of the second kind. Its bandwidth limit is computed by Θ-G: Taylor-expand ψ and p to second order about c, integrate against the Gaussian, use the vanishing of the odd moments and $\int w\, uu^\top du = \sigma^2 \kappa\, G_5^{-1}$, and divide by the degree. What survives at order σ^2 is $-\mathcal{L}_\star$.

(Q5) Retraction through the tower. Sampling the integral operator on $\mathcal{C}_M$ by the Riemann rule of Θ-J(1) returns the finite sum $(\Gamma W \Gamma) \otimes \mathsf{B}(\mathsf{A}_{57})$ with $O(\ell^2)$ quadrature error. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \; 0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta_{\text{ind}}^{(57)})]$ vanishes with $\nu \geq 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{57}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: Impossibility from linearity and unitarity.

Structure: The proof is two lines: linearity forces the cloner to act on superpositions term by term, giving $|\psi_1\rangle\,|\psi_1\rangle + |\psi_2\rangle\,|\psi_2\rangle$ rather than the product of sums. Cloning would require a nonlinear map, so no-cloning is a *consequence* of linearity, not an obstruction to it.

Reduction: The theorem follows directly from linearity.

Status: Exact linear — linearity is the reason for the impossibility.

8S status: Exact linear — linearity is the reason for the impossibility; preserved under the transform by Theorem C and Lemma T1.

Calculus status: Exact linear — linearity is the reason for the impossibility; carried to the continuum by Lemma D1, and the input was algebraic, so the differential content is entirely supplied by the transform.

58. Von Neumann Entropy

Origin (LA edition).

$$S(\rho) = -\operatorname{tr}(\rho \ln \rho) = -\sum_i \lambda_i \ln \lambda_i$$

Input (8S edition).

$$\mathcal{L}_{8S}\, \phi_k^{(58)} = \mu_k^{(58)} \phi_k^{(58)}, \qquad \mathcal{L}_{8S} = K_2 L_W + 2K_3 L_{\hat{H}}, \qquad 0 = \mu_1^{(58)} \leq \mu_2^{(58)} \leq \cdots \leq \mu_N^{(58)}$$

Calculus image.

$$-\frac{1}{2p^2} \nabla \cdot (p^2 G_5^{-1} \nabla \phi_k^{(58)}) = \mu_k^{(58)} \phi_k^{(58)}, \qquad \mu_2^{(58)} = \min_{\substack{\phi \neq 0 \\ \int \phi\, p^2 dc = 0}} \frac{\int_{\mathcal{C}} \|\nabla \phi\|^2_{G_5^{-1}}\, p^2\, dc}{\int_{\mathcal{C}} \phi^2\, p^2\, dc}$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T2**, so the calculus image is of class **D2** (Sturm–Liouville problem and Rayleigh quotient). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the Sturm–Liouville problem and Rayleigh quotient of Lemma D2, instantiated at the point $c_{58} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D2 needs **(R1)**, **(R2)** and the standing normalisation **(N)**. Discreteness of the spectrum uses compactness of $\mathcal{C}$ and the Rellich–Kondrachov embedding $H^1 \hookrightarrow L^2$; without compactness the spectrum could be continuous and μ_2 would not be attained. The min–max formula needs self-adjointness on $L^2(p^2 dc)$, which is Θ-C(2).

(Q4) Derivation. Θ-G supplies the operator by the Gaussian-moment expansion; Θ-C(1) converts it into the bilinear form by the divergence theorem, the boundary term vanishing under the Neumann condition the kernel limit imposes. Imposing the normalisation $\int \phi^2 p^2 = 1$ and taking the first variation of $\mathcal{E}$ produces the eigenvalue equation with μ as the Lagrange multiplier; Courant–Fischer then gives the displayed variational characterisation of μ_2 on the complement of the constants.

(Q5) Retraction through the tower. Discretising the Rayleigh quotient on $\mathcal{C}_M$ returns $x^\top \mathcal{L}_{8S} x / x^\top x$, whose minimiser over mean-zero x is the 8S μ_2; the gradient becomes a difference quotient with $O(\ell^2)$ error. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \;\, 0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta_{\text{ind}}^{(58)})]$ vanishes with $\nu \geq 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{58}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: A spectral functional of the density matrix.

Structure: Zero for pure (rank-one) states, maximal $\ln d$ for the maximally mixed $\rho = I/d$. Invariant under unitary evolution since the eigenvalues are, so entropy increase requires an open system. Entanglement entropy is S of a reduced density matrix.

Reduction: Entropy is a spectral functional.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T2.

Calculus status: Exact linear; carried to the continuum by Lemma D2, and the input was algebraic, so the differential content is entirely supplied by the transform.

Also in: Information Theory, Statistical Mechanics.

59. Decoherence Rate

Origin (LA edition).

$$\rho_{ij}(t) = \rho_{ij}(0) e^{-\Gamma_{ij} t}$$

Input (8S edition).

$$\mathcal{L}_{8S}\, \phi_k^{(59)} = \mu_k^{(59)} \phi_k^{(59)}, \qquad \mathcal{L}_{8S} = K_2 L_W + 2K_3 L_{\hat{H}}, \qquad 0 = \mu_1^{(59)} \leq \mu_2^{(59)} \leq \cdots \leq \mu_N^{(59)}$$

Calculus image.

$$-\frac{1}{2p^2} \nabla \cdot (p^2 G_5^{-1} \nabla \phi_k^{(59)}) = \mu_k^{(59)} \phi_k^{(59)}, \qquad \mu_2^{(59)} = \min_{\substack{\phi \neq 0 \\ \int \phi\, p^2 dc = 0}} \frac{\int_{\mathcal{C}} \|\nabla \phi\|^2_{G_5^{-1}}\, p^2\, dc}{\int_{\mathcal{C}} \phi^2\, p^2\, dc}$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T2**, so the calculus image is of class **D2** (Sturm–Liouville problem and Rayleigh quotient). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the Sturm–Liouville problem and Rayleigh quotient of Lemma D2, instantiated at the point $c_{59} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D2 needs **(R1)**, **(R2)** and the standing normalisation **(N)**. Discreteness of the spectrum uses compactness of $\mathcal{C}$ and the Rellich–Kondrachov embedding $H^1 \hookrightarrow L^2$; without compactness the spectrum could be continuous and μ_2 would not be attained. The min–max formula needs self-adjointness on $L^2(p^2 dc)$, which is Θ-C(2).

(Q4) Derivation. Θ-G supplies the operator by the Gaussian-moment expansion; Θ-C(1) converts it into the bilinear form by the divergence theorem, the boundary term vanishing under the Neumann condition the kernel limit imposes. Imposing the normalisation $\int \phi^2 p^2 = 1$ and taking the first variation of $\mathcal{E}$ produces the eigenvalue equation with μ as the Lagrange multiplier; Courant–Fischer then gives the displayed variational characterisation of μ_2 on the complement of the constants.

(Q5) Retraction through the tower. Discretising the Rayleigh quotient on $\mathcal{C}_M$ returns $x^\top \mathcal{L}_{8S} x / x^\top x$, whose minimiser over mean-zero x is the 8S μ_2; the gradient becomes a difference quotient with $O(\ell^2)$ error. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \;\, 0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta_{\text{ind}}^{(59)})]$ vanishes with $\nu \geq 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{59}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: Exponential decay of off-diagonal density matrix entries.

Structure: Decoherence is the density matrix becoming diagonal in the pointer basis — off-diagonals decaying while the diagonal is preserved. It is a rank-reducing, entropy-increasing process, and the pointer basis is selected by the system–environment coupling operator's eigenvectors.

Reduction: Decoherence is off-diagonal decay.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T2.

Calculus status: Exact linear; carried to the continuum by Lemma D2, and the input was algebraic, so the differential content is entirely supplied by the transform.

60. Lindblad Master Equation

Origin (LA edition).

$$\dot{\rho} = -\frac{i}{\hbar}[H, \rho] + \sum_k \left(L_k \rho L_k^\dagger - \tfrac{1}{2}\{L_k^\dagger L_k, \rho\} \right)$$

Input (8S edition).

$$\operatorname{rank}(C_5) = \operatorname{rank}(C_4) + \mathbf{1}[\eta_{\text{ind}}^{(60)} > 0], \qquad \dim\ker \mathcal{L}_{8S} = \#\operatorname{comp}(W + \hat{H})$$

Calculus image.

$$\dim\ker \mathcal{L}_\star = \dim H^0(\mathcal{C}) = \#\{\text{components}\}, \qquad \operatorname{ind} \mathcal{L}_\star = 0, \qquad \det \operatorname{Gr}(C_4, e_{60}) \neq 0 \iff \text{rank increases}$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T5**, so the calculus image is of class **D5** (kernel dimension, index, and H^0). The originating equation is first order — it already carries a derivative or an integral. The input already carries a first-order differential or integral operator. On that part the transform acts as the identity — a derivative is already a calculus object — and only the coupling and the gate are taken to the continuum. The regularity demanded of the input is correspondingly one order lower.

(Q2) Image. The displayed relation is the kernel dimension, index, and H^0 of Lemma D5, instantiated at the point $c_{60} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D5 needs $\mathcal{L}_\star$ self-adjoint with compact resolvent, which is Θ-G under **(R1)**–**(R2)**; the identification of ker with locally constant functions needs $G_5 \succ 0$ and $p > 0$, both standing. The index vanishes because the operator is self-adjoint, so coker ≅ ker.

(Q4) Derivation. Set $\mathcal{E}[\phi] = 0$ in Θ-C(1). Since $G_5^{-1} \succ 0$ and $p > 0$, this forces $\nabla\phi = 0$ almost everywhere, hence ϕ locally constant, hence $\ker \mathcal{L}_\star$ spanned by the indicator functions of the connected components — which is precisely H^0. Self-adjointness gives coker ≅ ker and therefore index zero.

(Q5) Retraction through the tower. On the lattice $\dim\ker \mathcal{L}_\star$ becomes $\dim\ker \mathcal{L}_{8S}$, the component count of the coupling graph, and the Gram condition becomes the 8S rank increment. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \; 0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta_{\text{ind}}^{(60)})]$ vanishes with $\nu \geq 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{60}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: The most general linear generator preserving trace and positivity.

Structure: Linear in ρ, so the whole open-system evolution is a linear map on the d^2-dimensional space of density matrices — vectorised, it is a $d^2 \times d^2$ matrix. Complete positivity is the structural constraint, and the steady state is a null vector of that superoperator. Even dissipation and measurement remain exactly linear.

Reduction: Open evolution is a linear superoperator.

Status: Exact linear. The chapter closes as it began: nothing here is anything but linear algebra.

8S status: Exact linear. The chapter closes as it began: nothing here is anything but linear algebra; preserved under the transform by Theorem C and Lemma T5.

Calculus status: Exact linear. The chapter closes as it began: nothing here is anything but linear algebra; carried to the continuum by Lemma D5, and the input already carried a first-order operator, which the transform leaves in place.